MEDISMART: SMART PHARMACY & PRESCRIPTION SYSTEM

A MINI PROJECT REPORT

Submitted by

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MLM25MCA-2020

To

the APJ Abdul Kalam Technological University in

Partial fulfillment of the requirements for the

award of the degree

of

MASTER OF COMPUTER APPLICATIONS

Under the Guidance of

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DEPARTMENT OF COMPUTER APPLICATIONS

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR

(Affiliated to APJ Abdul Kalam Technological University)

OCTOBER 2026

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DEPARTMENT OF COMPUTER APPLICATIONS

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To become a centre of excellence in computer applications,competent in the global ecosystem with technical knowledge,innovation with a sense of social commitment.

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- **PSO I**: Apply advanced technologies through innovations to enhance the efficiency of design and development.
- **PSO II**: Apply the principles of computing to analyze, design and implement sustainable solutions for real world challenges.

MAPPING OF PO-PSO-SDG

1. MAPPING WITH PROGRAM OUTCOMES (POs):

SL.NO	POs ADDRESSED	RELEVANCE TO PROJECT
1	PO1	The project applies programming knowledge, database Concepts and software development principles to develop a smart pharmacy management system.
2	PO2	The project identifies and addresses problems in traditional pharmacy operations, prescription processing and medicine availability management
3	PO3	A web-based solution is developed to provide online medicine ordering, prescription verification and efficient pharmacy management.
4	PO4	Modern technologies such as Python, Flask, SQLite, OCR, TrOCR and Google Gemini API are utilized for system development and prescription processing.
5	PO6	Project activities such as module development, resource utilization, task planning and testing are organized systematically.
6	PO8	The project encourages continuous learning and adaptation to emerging technologies such as AI-based prescription processing and OCR.

LIST OF PROGRAM OUTCOMES (POs):

PO1 (Foundation Knowledge): Apply knowledge of programming, database concepts and software engineering principles to develop effective software solutions.

PO2 (Problem Analysis): Identify, analyze and understand real-world problems and develop suitable solutions based on user requirements.

PO3 (Development of Solutions): Design and develop innovative software applications to solve practical problems using appropriate technologies.

PO4 (Modern Tool Usage): Select and apply modern computational tools and technologies for software development, implementation and testing.

PO5 (Individual and Teamwork): Work effectively as an individual or as a member of a team during software development, implementation and testing activities.

PO6 (Project Management and Finance): Apply project planning, resource management and time management principles for successful project development.

PO7 (Ethics): Follow professional ethics and security practices to protect user information, maintain data privacy and ensure responsible system usage.

PO8 (Life-long Learning): Develop continuous learning skills and adapt to emerging technologies and modern software development practices.

2. MAPPING WITH PROGRAM SPECIFIC OUTCOMES (PSOs):

SL.NO	PSOs ADDRESSED	RELEVANCE TO PROJECT
1	**PSO 1**	Applies AI, OCR, image processing and modern computational technologies to develop an intelligent prescription verification system.
2	**PSO 2**	Applies computing principles and database management techniques to provide an efficient solution for medicine ordering, pharmacy management and prescription processing.

LIST OF PROGRAM SPECIFIC OUTCOMES (PSOs):

PSO1 (Advanced Technologies and Innovation): Apply advanced computing technologies, programming concepts and innovative approaches to design and develop intelligent software solutions for real-world problems.

PSO2 (Computing and Sustainable Solutions): Apply computing knowledge, database technologies and modern computational tools to develop efficient, user-oriented solutions addressing practical and technological needs.

3. MAPPING WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs):

SDG NO	SDGs ADDRESSED	RELEVANCE TO PROJECT
SDG 3	Good Health and Well-being	Provides convenient access to medicine information and supports prescription verification through a digital pharmacy platform.
SDG 9	Industry, Innovation and Infrastructure	Utilizes AI, OCR and modern web technologies to develop an innovative and efficient pharmacy management system.
SDG 12	Responsible Consumption and Production	Supports efficient medicine stock and expiry-date management, helping to reduce medicine wastage.

SUSTAINABLE DEVELOPMENT GOALS (SDGs):

SDG 1 – No Poverty-End poverty in all its forms everywhere.

SDG 2 – Zero Hunger-End hunger, achieve food security and improved nutrition, and promote sustainable agriculture.

SDG 3 – Good Health and Well-Being-Ensure healthy lives and promote well-being for all at all ages.

SDG 4 – Quality Education-Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

SDG 5 – Gender Equality-Achieve gender equality and empower all women and girls.

SDG 6 – Clean Water and Sanitation-Ensure availability and sustainable management of water and sanitation for all.

SDG 7 – Affordable and Clean Energy-Ensure access to affordable, reliable, sustainable, and modern energy for all.

SDG 8 – Decent Work and Economic Growth-Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all.

SDG 9 – Industry, Innovation, and Infrastructure-Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.

SDG 10 – Reduced Inequality-Reduce inequality within and among countries.

SDG 11 – Sustainable Cities and Communities-Make cities and human settlements inclusive, safe, resilient, and sustainable.

SDG 12 – Responsible Consumption and Production-Ensure sustainable consumption and production patterns.

SDG 13 – Climate Action-Take urgent action to combat climate change and its impacts.

SDG 14 – Life Below Water-Conserve and sustainably use the oceans, seas, and marine resources.

SDG 15 – Life on Land -Protect, restore, and promote sustainable use of terrestrial ecosystems, manage forests sustainably, combat desertification and halt biodiversity loss.

SDG 16 – Peace, Justice, and Strong Institutions- Promote peaceful and inclusive societies, provide access to justice for all, and build effective, accountable, and inclusive institutions.

SDG 17 – Partnerships for the Goals -Strengthen the means of implementation and revitalize the global partnership for sustainable development.

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR

DEPARTMENT OF COMPUTER APPLICATIONS

OCTOBER 2026



DECLARATION

I hereby certify that the work presented in the project entitled "MEDISMART – SMART PHARMACY AND PRESCRIPTION SYSTEM" submitted to the DEPARTMENT OF COMPUTER APPLICATIONS is an authentic record of my own work carried out under the supervision of Mr. ELDHOSE K PAUL, ASSOCIATE PROFESSOR. This work has not been submitted to any other institution or university for the award of any other degree. The report has been checked for plagiarism by the college and the similarity index is within the permissible limit set by the college.

Date:

Place:

Name & Signature of Student

ANJANA JAYAN

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR

DEPARTMENT OF COMPUTER APPLICATIONS

OCTOBER 2026

CERTIFICATE

Certified that the report entitled **"MEDISMART – SMART PHARMACY PRESCRIPTION SYSTEM"** submitted by **ANJANA JAYAN (MLM25MCA-2020)** to APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the **Degree of Master of Computer Applications** is a bonafide record of the MINI PROJECT work carried out by the student under our guidance and supervision during the academic year 2025–2027.

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ACKNOWLEDGEMENT

I am greatly indebted to the authorities of Mangalam College of Engineering for providing the necessary facilities to successfully complete my Project on the topic **"Medismart- Smart Pharmacy & Prescription System"**

I express my sincere thanks to **Dr. Vinodh P Vijayan**, Principal, Mangalam College of Engineering for providing the facilities to complete my Project successfully.

I thank and express my sincere gratitude to **Ms. Divya S B,** HOD, Department of Computer Applications, Mangalam College of Engineering, for her invaluable help and support which helped me a lot in successfully completing this Project.

I express my gratitude to my Internal Guide, **Mr. Eldhose K Paul**, Associate professor, Department of Computer Applications for the valuable suggestions and encouragement which helped in the successful completion of my Project.

Furthermore, I would like to acknowledge with much appreciation the crucial role of the faculties especially Project coordinator, **Ms. Athira R Kurup**, Department of Computer Applications, Mangalam College of Engineering, who gave the permission to use all the required equipment and the necessary resources to complete the presentation & report.

Finally, I would like to express my heartfelt thanks to my parents who were very supportive both financially and mentally and for their encouragement to achieve my goal.

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ABSTRACT

MediSmart – Smart Pharmacy & Prescription System is a web-based application developed to connect customers and pharmacies through a single digital platform. It addresses common problems in traditional pharmacy management, including manual stock maintenance, inaccurate expiry tracking, delays in prescription verification, and limited online access to pharmacy services. The system enables customers to search and order medicines, upload prescriptions for AI-assisted verification, make payments, and track orders. Pharmacy administrators can manage users, medicines, categories, stock, and prescriptions through a dedicated dashboard.

The system follows a layered architecture using HTML, CSS, JavaScript, Bootstrap, Flask, Python, and SQLite. Tesseract OCR, TrOCR, and the Google Gemini API are intended to support prescription reading and verification, while a rule-based, NLP-supported chatbot answers common customer queries.

The project is documented using Data Flow Diagrams, Use Case, Class, State/Activity, Sequence, Entity-Relationship, and System Architecture diagrams. At the current development stage, approximately 80% of the front-end has been completed, and key interfaces have been functionally tested. Full integration with live AI services, completion of the administrator dashboard, payment gateway integration, and delivery logistics remain future work.

MediSmart aims to improve the accessibility, accuracy, and efficiency of pharmacy services through digital management and AI-assisted processing. It can help reduce manual workload, improve medicine record maintenance, support prescription review, and monitor expiry dates to minimize medicine wastage. By providing customers with convenient online medicine services and administrators with centralized management tools, the system establishes a foundation for modern pharmacy operations. Future enhancements can make the platform more reliable and scalable through complete AI integration, secure online payments, and efficient delivery management, contributing to better healthcare accessibility and improved pharmacy service quality.

Keywords: MediSmart, Smart Pharmacy, AI Prescription Verification, Optical Character Recognition (OCR), Medicine Management, Prescription Processing, Medicine Ordering, Stock Management, Expiry Tracking, Artificial Intelligence, Healthcare Management System.

Mapping with Sustainable Development Goals	SDG 3 – Good Health and Well-being SDG 8 – Decent Work and Economic Growth SDG 9 – Industry, Innovation, and Infrastructure SDG 12 – Reduced Inequalities

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LIST OF ABBREVIATIONS

ABBREVIATION		FULL FORM
AI	-	Artificial Intelligence
API	-	Application Programming Interface
CNN	-	Convolutional Neural Network
DFD	-	Data Flow Diagram
ER	-	Entity-Relationship
HTML	-	HyperText Markup Language
CSS	-	Cascading Style Sheet
LSTM	-	Long Short-Term Memory
NLP	-	Natural Language Processing
MCA	-	Master of Computer Applications
OCR	-	Optical Character Recognition
JS	-	Javascript
TrOCR	-	Transformer-based Optical Character Recognition
SQLite	-	Structured Query Language-Lite
UI	-	User Interface

CHAPTER 1
INTRODUCTION

1.1 Background

Community pharmacies play an important role in healthcare by providing medicines and ensuring that patients receive the correct treatment. However, many pharmacies still depend on manual or semi-digital methods for tasks such as prescription verification, stock management, purchase records, and expiry tracking. These methods can lead to errors, delays, and difficulties in managing pharmacy operations efficiently.With the growth of internet and cloud-based technologies, customers increasingly expect convenient services such as checking medicine availability, uploading prescriptions, placing orders, and tracking purchases online. At the same time, pharmacies require effective tools to manage medicines, inventory, prescriptions, and orders in an organized manner.Artificial Intelligence technologies such as Optical Character Recognition (OCR) and Natural Language Processing (NLP) can assist in reading and understanding prescription information, reducing manual effort and supporting faster verification. Human review can still be maintained to ensure the accuracy and safety of prescription approval.

MediSmart – Smart Pharmacy & Prescription System is a web-based application developed to address these needs. It provides customers with facilities to search for medicines, upload prescriptions, place orders, and track their orders. It also provides pharmacy administrators with a centralized system for managing medicines, categories, stock, prescriptions, and orders. The system aims to make pharmacy operations more organized, convenient, and efficient while maintaining prescription verification as an important safety step.

1.2 Motivation

The motivation behind MediSmart arises from the common challenges faced in traditional management, medicine details and expiry-date tracking. These issues may result in medicine shortages or wastage of medicines that have passed their expiry dates, affecting both pharmacy operations and patient convenience.

Another important motivation is the time required for prescription verification. Pharmacists need to read prescriptions, identify the prescribed medicines and check their availability before processing orders. MediSmart aims to simplify this process through AI-assisted prescription reading using OCR, TrOCR and Google Gemini API, while allowing manual verification whenever necessary to maintain accuracy and safety.

The increasing demand for convenient healthcare services also motivates the development of this system. Customers can search for medicines, check availability, upload prescriptions and track their orders through an online platform without making repeated visits to pharmacies. Furthermore, MediSmart provides an opportunity to integrate modern web development technologies with AI-based tools to create a practical pharmacy management solution. The project aims to reduce manual effort, improve efficiency in pharmacy operations and provide a more convenient experience for both customers and pharmacists.

1.3 Scope & Objectives

The scope of MediSmart focuses on developing a web-based Smart Pharmacy and Prescription System that simplifies pharmacy operations and improves customer convenience. The system provides two main interfaces: a customer interface and an administrator interface. Customers can register, log in, search and browse medicines by name or category, check medicine availability, upload prescriptions for AI-assisted verification, place orders, make online payments and track their order status.The administrator interface allows pharmacy staff to manage users, medicines, categories, stock details, prescriptions and customer orders. It also supports prescription review and approval, along with access to payment-related reports. The system integrates AI technologies such as OCR, TrOCR and Google Gemini API to assist in reading and processing uploaded prescriptions. When automated verification is inconclusive, the prescription can be forwarded for manual review by the administrator.

The main objective of MediSmart is to provide an efficient and convenient digital platform for pharmacy services while reducing manual work and improving the accuracy of pharmacy management. It aims to simplify medicine ordering, support reliable prescription verification, maintain proper stock and expiry-date records, reduce medicine wastage and provide customers with an improved online purchasing experience. The system also includes chatbot-based customer support to assist users with common queries and uses a modular architecture that supports future maintenance and development.

1.4 Problem Statement

Pharmacies that continue to rely on manual or partially digital methods for managing medicines, stock and prescriptions face several recurring problems. Manual entry of medicine details and stock counts can cause data inaccuracy, which in turn leads to incorrect stock levels being displayed or assumed, resulting either in lost sales when a medicine is actually available but shown as out of stock, or in customer dissatisfaction when

Verifying a prescription without any digital assistance takes considerable time, especially during busy hours, because the pharmacist must read and interpret the prescription manually before dispensing medicines. This slows down service and increases the chance of misreading a medicine name, dosage or quantity. In addition, tracking the expiry dates of hundreds of medicine items on paper or in simple spreadsheets is difficult, and expired stock is often discovered only when a customer complains or during a physical audit, leading to medicine wastage and potential health risk if not caught in time.

A further problem is the absence of a direct digital connection between customers and pharmacies. Customers cannot check medicine availability before visiting the store, cannot upload a prescription in advance to save time, and have no structured way of tracking the status of an order once it has been placed. The problem addressed by this project, therefore, is the need for an integrated web-based system that connects customers and pharmacies on one platform, automates the first level of prescription verification using AI, and gives pharmacy staff the tools to manage stock, expiry, prescriptions, orders and payments accurately and efficiently.

In summary, the central problem is one of fragmentation and manual dependency: the pharmacy's inventory records, its prescription verification process and its customer-facing ordering channel are all handled separately, largely by hand, with no shared digital system tying them together. MediSmart is designed to resolve this fragmentation by bringing these three concerns — inventory, prescription verification and customer ordering — onto a single, coherent platform.

CHAPTER 2
REQUIREMENT AND ANALYSIS

2.1 Existing System and Problem Statement

The existing pharmacy system, as observed in most small and medium sized outlets, uses manual and basic digital methods to manage stock and records. Medicine details, stock quantities and expiry dates are commonly tracked on paper registers or in simple spreadsheets that are updated by hand whenever stock is received or sold. Prescription verification is performed entirely by the pharmacist reading the physical prescription, and customers are required to visit the pharmacy directly to place an order and collect their medicines. While the existing system works because of the effort and experience of pharmacy staff, it depends almost entirely on human attention and does not provide any digital record that customers can access remotely.

The key limitations observed in the existing system are summarised below:

- Manual records of stock, expiry dates and medicine availability are maintained on paper or in isolated spreadsheets, making it difficult to get a consolidated, real-time view of inventory.
- Human errors during manual recording of medicine details can lead to mismatched stock counts and incorrect billing.
- Expired medicines are not tracked systematically, which increases the risk of medicine wastage and, if undetected, of dispensing expired stock.
- Prescriptions are verified only by manual reading, with no authenticity check or digital record, which can slow down service during busy hours and makes it difficult to review prescription history later.
- There is no direct online channel connecting customers and pharmacies, so customers cannot check medicine availability, upload prescriptions in advance, or track an order without physically visiting or calling the pharmacy.

These limitations compound one another in practice. A pharmacy that cannot rely on accurate stock records is also less able to plan purchasing decisions confidently, which can lead to either overstocking (tying up capital and increasing the risk of expiry-related wastage) or understocking (leading to lost sales and customer dissatisfaction). Similarly, a

pharmacy without a digital prescription record has no easy way to look back at a customer's prescription history, which can be useful both for customer service and for identifying patterns such as recurring prescriptions for chronic conditions. The absence of a shared digital thread connecting these day-to-day activities is the central gap that MediSmart is designed to close.

2.2 Solution Overview

MediSmart is a web-based smart pharmacy system developed to simplify and organize pharmacy-related activities through a single platform. The system allows customers to search for medicines, view available products, upload prescriptions, place orders, make payments, and track their orders. By providing these services online, the system reduces the dependency on manual processes and makes pharmacy services more convenient for customers.

One of the main features of MediSmart is AI-assisted prescription verification. Customers can upload prescription images, which are processed using OCR and AI-based techniques to extract relevant information. The system assists in analysing the prescription and identifying the required medicines. If the prescription is unclear or requires additional verification, it can be referred to the administrator for manual checking, thereby combining automated processing with human supervision.

The system also provides pharmacy management features for administrators. Administrators can manage medicines, categories, stock, expiry information, users, prescriptions, and orders through the system. Inventory and expiry management helps maintain updated stock information and monitor medicines approaching their expiry date. The ordering and payment modules also allow customers to complete purchases and view the status of their orders through the platform.

In addition to the core pharmacy functions, MediSmart provides chatbot-based support and reporting facilities. The chatbot can assist customers with common queries, while reports provide administrators with useful information related to medicines, orders, users, prescriptions, and inventory. Overall, the proposed system integrates customer services and pharmacy management functions into one organized platform, helping to improve efficiency and reduce manual effort.

2.3 Feasibility Study

The MediSmart system is feasible in terms of technology, cost, and operation. It uses commonly available technologies such as HTML, CSS, JavaScript, Bootstrap, Python, Flask, and SQLite, along with OCR and AI technologies for prescription processing. Since most of the technologies are open-source or freely available, the development cost can be kept low. The system is also easy to operate, allowing customers to search medicines, upload prescriptions, and place orders, while administrators can manage medicines, stock, prescriptions, and orders efficiently. Therefore, the proposed system is practical and suitable for implementation.

2.3.1 Technical Feasibility

MediSmart is built entirely using widely available, well documented and free-to-use technologies, which makes it technically feasible to implement within an academic project timeline. The front end uses HTML, CSS, JavaScript and the Bootstrap framework to build a responsive user interface. The back end is developed using Python with the Flask framework, which is lightweight and well suited to a project of this scale, and the data is stored in an SQLite database, which does not require a separate database server to be installed and administered. AI-based prescription reading is supported using Tesseract OCR, TrOCR and the Google Gemini API, all of which provide accessible libraries or APIs that can be integrated into a Flask application without specialised infrastructure. Since all of these tools are open source or offered with a free tier, and since they integrate well with each other, the system is technically feasible to develop, test and maintain.

2.3.2 Economic Feasibility

The technologies chosen for MediSmart are open-source or available at low or no cost for a project of this scale, which keeps the overall development cost low. There is no license fee for Flask, SQLite, Bootstrap or Tesseract OCR, and the Google Gemini API offers a usage tier suitable for prototyping and small-scale deployment. Because the system does not require expensive proprietary software or specialised hardware, it is economically feasible for a small or medium sized pharmacy to adopt a system built on this stack, and the cost of development within the academic project remains within the resources available to a student developer.

2.3.3 Operational Feasibility

Operationally, MediSmart is designed with a simple, intuitive, and user-friendly interface so that both customers and pharmacy administrators, who may not have advanced technical knowledge, can use the system with minimal training. Customers can interact with the system in the same way they would use any modern e-commerce website by registering, logging in, searching for medicines, checking availability, uploading prescriptions, adding medicines to the cart, making payments, and tracking their orders. The organized layout and straightforward navigation help users access the required features easily and complete their activities without unnecessary difficulties.

For pharmacy administrators, MediSmart provides a centralized dashboard that helps manage daily pharmacy operations more efficiently. Administrators can add, update, and remove medicine details, manage medicine categories, monitor stock availability, review uploaded prescriptions, process customer orders, and track medicine expiry dates. These functions help reduce the need for maintaining separate manual records and make it easier to access and update information whenever required. The system also helps pharmacy staff identify medicines with low stock levels and monitor expiry information, supporting better inventory management and reducing the possibility of errors caused by manual record-keeping.

MediSmart also improves communication and coordination between customers andpharmacy staff by providing a structured process for prescription submission, verification, and order processing. Customers can check medicine information and order status without always needing to contact the pharmacy directly. The AI-based prescription verification feature is intended to assist the prescription review process by extracting and analyzing relevant information from uploaded prescription images. However, prescriptions requiring clarification or professional judgment can still be reviewed by an authorized pharmacy administrator to ensure appropriate handling.

CHAPTER 3
LITERATURE REVIEW

3.1 AI-Based Prescription Reader (2026)

Amal Mariya Sibi & Bijimol T.K., National Conference on Emerging Computing Applications (NCECA 2026).

This work proposes an AI-based prescription reader that applies image processing, OCR and a Convolutional Neural Network (CNN) together with NLP techniques to identify medicines from a prescription image. The pipeline consists of preprocessing the prescription image, extracting the text through OCR, and then identifying the specific medicine name from the extracted text. The authors report a recognition accuracy of 95.4% for medicine identification using this pipeline.

Key Aspects

- Follows a three-stage pipeline of preprocessing, text extraction and medicine identification, which closely matches the AI Processing Module planned for MediSmart.
- Combines CNN-based image analysis with NLP for interpreting extracted prescription text.
- Reports a high recognition accuracy (95.4%), demonstrating that OCR-based prescription reading can achieve reliable results when properly preprocessed.

This paper is one of the closest matches to the AI Prescription Verification feature of MediSmart, as it validates that a preprocessing, OCR and identification pipeline of the kind planned for this project can achieve high recognition accuracy in a related pharmacy context. The reported 95.4% figure was obtained on the dataset used in that specific study; it is cited here as an external benchmark rather than as a result achieved by MediSmart itself, consistent with the position taken in Section 8.1.2 that MediSmart's own accuracy has not yet been measured.

3.2 Medicine Management System

Ruth G. Luciano et al., International Journal of Multidisciplinary: Applied Business and Education Research (IJEME)

This paper presents a Medicine Management System developed using HTML, PHP, XAMPP and a MariaDB database, following an Agile development methodology.

The system supports online medicine requisition, prescription upload, user authentication and database-driven management of medicine records. The study evaluated the system primarily in terms of usability, examining how easily end users were able to navigate the requisition and prescription upload process.

Key Aspects

- Implements digital requisition and prescription upload as core functions of the system.
- Uses an Agile methodology for iterative development, similar to the modular approach followed in MediSmart.
- Evaluation is centred on system usability rather than algorithmic accuracy.

The paper is relevant to the requirement analysis and module design of MediSmart, since it confirms that digital requisition, prescription upload and authentication are commonly treated as essential modules of a medicine management system, mirroring the User, Prescription and Order modules implemented in this project. Because its evaluation is usability-focused rather than accuracy-focused, it is best read as supporting evidence for MediSmart's module boundaries (Section 5.1) rather than for the AI verification component specifically.

3.3 PharmaGo: An Online Pharmaceutical Ordering Platform

Ranula Gihara Gamage et al., IEEE International Conference on Advancements in Computing (ICAC).

PharmaGo is an online pharmaceutical ordering platform that combines prescription reading with pharmacy recommendation and demand prediction. The system uses Optical Character Recognition (OCR) and image processing techniques to extract medicine names from an uploaded prescription image, and applies a Long Short-Term Memory (LSTM) based model together with an AI chatbot to assist users during the ordering process. The reported focus of the study's evaluation was on the prediction performance of the demand forecasting component.

Key Aspects

- Combines OCR-based prescription reading with an AI chatbot for a conversational ordering experience.
- Uses an LSTM model to predict medicine demand, which can help pharmacies plan stock.
- Provides pharmacy recommendations to the user in addition to allowing online

Ordering. This paper is directly relevant to MediSmart because it validates the approach of combining OCR-based prescription reading with an online ordering workflow and a chatbot for user assistance, both of which are core features of the proposed system. As the paper's own evaluation centres on demand-prediction performance rather than prescription-reading accuracy, it does not by itself establish how accurately its OCR component reads prescriptions, which is why the accuracy figure reported in Section 3.3 is treated separately below.

3.4 Patient Satisfaction with Online Pharmacy Services

Etidal-Mihaela Manoliu-Hamwi et al., IEEE International Conference on E-Health and Bioengineering (EHB 2022).

This study evaluates patient satisfaction with online pharmacy services using a structured questionnaire, with responses analysed statistically using SPSS and the Pearson Chi-square test. The evaluation measured satisfaction percentages and statistical significance (p-value) associated with different aspects of the online pharmacy experience, and the study reported a high overall level of patient satisfaction with online pharmacy services.

Key Aspects

- Uses a structured questionnaire and statistical hypothesis testing (Pearson Chi-square) to evaluate user satisfaction.
- Focuses specifically on the patient/customer perspective of online pharmacy services, rather than on the technical implementation.
- Reports high overall satisfaction, supporting the broader case for digitising pharmacy services.
- Uses a structured questionnaire and Pearson Chi-square test to evaluate patient satisfaction.
- Reports high overall satisfaction, supporting the adoption of digital pharmacy services.

This paper supports the motivation for MediSmart by providing evidence that patients respond positively to online pharmacy services in general, reinforcing the rationale for building an online ordering and prescription platform. Its findings relate to online pharmacy services broadly rather than to any AI-assisted verification feature specifically, so it is used in this report as motivational and comparative evidence (Section 8.1.2) rather than as technical guidance for the AI Processing Module.

3.5 PharmAI (2025)

Roshni Patane & Ramesh D. Jadhav.

PharmAI applies Artificial Intelligence and predictive analytics, modelled using UML, to manage pharmacy inventory, billing and reporting, and to predict demand for medicines. The study evaluates the system in terms of overall system efficiency and reports that the approach improves inventory management and supports better decision-making within the pharmacy.

Key Aspects

- Applies predictive analytics to inventory, billing and reporting, similar to the stock management and reporting objectives of MediSmart.
- Uses UML-based modelling for system design, consistent with the DFD, use case, class and sequence diagrams prepared for MediSmart.
- Demonstrates that AI-assisted analytics can improve inventory decision-making in a pharmacy setting.

This paper is relevant to the Stock Management and Reports & Analytics aspects of MediSmart, confirming that AI-assisted inventory and reporting functions are an active and beneficial area of research for pharmacy management systems. Since the supplied source material does not include the specific efficiency figures reported in this study, this report cites it for its qualitative conclusion only and does not restate any numeric result beyond what was provided.

3.6 Image-Based Extraction of Prescription Information using OCR-Tesseract (2024)

This study focuses on extracting medical information from prescription images using Optical Character Recognition (OCR) technology, particularly Tesseract OCR. The system processes prescription images and converts the text into a machine-readable format, helping identify medicine names and other prescription details. Image preprocessing techniques can Improve text clarity and support more accurate recognition. This approach reduces the need for manual data entry and helps digitize prescription information for healthcare applications. It is relevant to the MediSmart project because it demonstrates how OCR technology can support prescription reading and automated medicine information extraction.

Key Aspects

- Uses Tesseract OCR to extract text from prescription images.
- Converts handwritten or printed prescription content into machine-readable text, depending on image quality and handwriting legibility.
- Applies image preprocessing to improve text recognition.
- Supports the extraction of medicine names and prescription details.
- Reduces manual data entry and helps digitize prescription information.
- Provides a foundation for AI-assisted prescription processing in MediSmart.

3.7 AI-Enhanced Pharmacy Inventory Management System (2026)

The paper titled AI-Enhanced Pharmacy Inventory Management System was published in the IEEE International Conference on Emerging Trends and Innovations in ICT (ICEI), 2026. This study presents an AI-based approach to improving pharmacy inventory management and reducing problems associated with manual stock handling. It combines machine learning, Natural Language Processing (NLP), and semantic search to support medicine classification, inventory control, and user interaction. The system also incorporates inventory prioritization and a biomedical chatbot to provide medicine-related information. These features aim to improve pharmacy efficiency, simplify inventory operations, and support better decision-making. This study is relevant to the MediSmart project because it focuses on medicine stock management, expiry tracking, and chatbot functionality. The proposed approach provides a useful reference for integrating AI technologies into pharmacy management systems.

Key Aspects

- AI-Based Inventory Management: Uses artificial intelligence to support pharmacy inventory operations.
- Stock Control: Helps manage medicine availability and reduce stock-related problems..
- NLP and Semantic Search: Supports intelligent searching and improved user interaction.
- Biomedical Chatbot: Provides assistance with medicine-related information.
- Relevance to MediSmart: Supports the development of the stock management, expiry tracking, and chatbot modules.

3.8 Prospective Systematic Risk Analysis of the Digital Technology Use Within Pharmaceutical Care (2024)

This study investigates potential risks associated with the use of digital technologies in pharmaceutical care. The researchers applied Failure Mode and Effects Analysis (FMEA) to identify, evaluate, and prioritize risks related to telepharmacy, mobile health applications, artificial intelligence, and software systems. The study identified risks involving incomplete data entry, identity fraud, implementation errors, and technological limitations. It also proposed corrective measures to reduce these risks and improve patient safety. The research emphasizes that digital pharmacy systems must be designed with appropriate security controls, reliable data handling, and mechanisms for identifying errors. This study is relevant to MediSmart because the platform processes user information, prescription images, medicine details, and online orders. Its findings highlight the importance of protecting sensitive information and ensuring that AI-based prescription verification is supported by appropriate human review.

Key Aspects

- Identification and assessment of digital healthcare risks.
- Data security and identity protection.
- Reduction of software and data-processing errors.
- Relevance to MediSmart's prescription verification and user data security.

3.9 Online Pharmacies Selling Prescription Drugs: Systematic Review (2022)

This systematic review examines online pharmacies that sell prescription medicines, focusing on their characteristics, the quality of medicines available, and the behaviour of consumers purchasing medicines online. The researchers reviewed 46 eligible articles to understand the benefits and risks associated with online pharmaceutical services. The review identified concerns related to unregulated online pharmacies, counterfeit or poor-quality medicines, and the purchase of prescription drugs without appropriate professional supervision. It highlights the importance of pharmacy verification, regulatory compliance, and patient safety in online medicine transactions. This study is relevant to MediSmart because it emphasizes the need for a reliable online pharmacy platform that supports prescription verification before eligible medicines are ordered. It also reinforces the importance of maintaining trustworthy medicine information and providing a secure purchasing process.

3.10 Artificial Intelligence in Pharmacy Practice: A Review of Current Applications and Future Perspectives

This review explores the use of Artificial Intelligence (AI) in pharmacy practice, including prescription processing, medication management, and automation of pharmacy operations. AI can improve efficiency, support medicine information analysis, and reduce manual errors. However, data accuracy, privacy, and professional supervision remain important. This study is relevant to MediSmart because it supports the integration of AI-based prescription verification and medicine management.

Key Aspects

- AI-based prescription processing.
- Improved medicine management.
- Reduction of manual errors.
- Data security and patient safety.
- Relevance to MediSmart's AI verification module.

CHAPTER 4
DIAGRAMS

4.1 Data Flow Diagrams (DFD)

A Data Flow Diagram (DFD) is a graphical representation used to describe how data moves through a system. It explains the flow of information between external entities, system processes, and data stores. In the MediSmart system, DFDs are used to represent the interaction between users, administrators, prescription processing, medicine management, inventory, order management, payment processing, and order tracking.

4.1.1 DFD Level 0

DFD Level 0, also known as the Context Diagram, provides a high-level view of the MediSmart system. The entire application is represented as a single process that interacts with three external entities: the User, Administrator, and Payment Service. The User provides login details, prescription images, medicine and order information, while receiving prescription results, medicine details, order confirmation, payment status, and tracking information. The Administrator manages users, medicines, inventory, and orders, and receives the required management information and reports. The Payment Service processes payment requests and returns the transaction status to MediSmart.

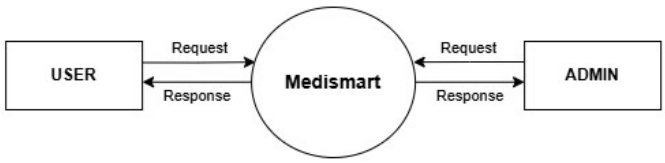


Figure 4.1.1: Level 0 DFD

4.1.2 DFD Level 1: User

The Level 1 DFD for the User module provides a detailed representation of the major functions performed by the user in the MediSmart system. The single process shown in the Level 0 DFD is decomposed into six main sub-processes: Profile Management, Search Medicine, Upload Prescription, Order Management, Payment, and Order Tracking. These processes describe how the user interacts with different features of the system.

The Profile Management process handles user registration, login, and profile-related information. Search Medicine allows the user to search for medicines and view their details and availability. Through Upload Prescription, the user can submit a prescription image for processing and verification. Order Management handles activities such as adding medicines to the cart, placing orders, and receiving order-related information.

The Payment process manages payment requests and stores the corresponding payment details in the Payments data store. Finally, Order Tracking allows the user to check the current status of their order using the Track data store. Each process communicates with its corresponding data store, such as Users, Medicines, Prescriptions, Orders, Payments, and Track, ensuring that user requests are processed and the required responses are provided. Thus, the Level 1 DFD gives a clearer view of how the different user-facing functions of MediSmart interact with the system's data.

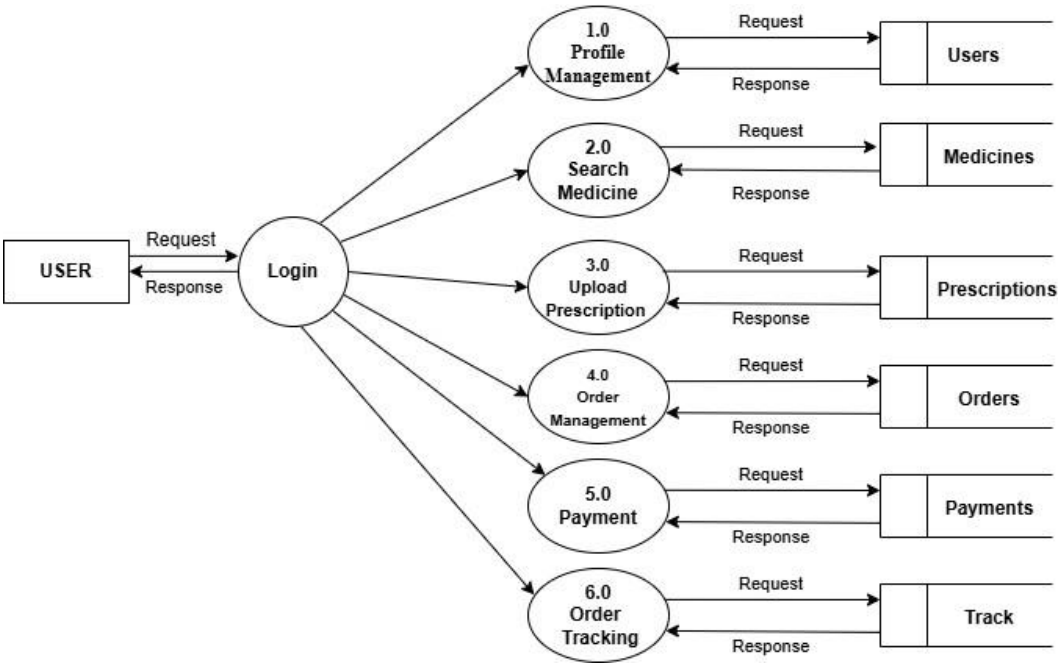


Figure 4.1.2: Level 1 DFD: User

4.1.3 Level 1 DFD: Admin

The Level 1 DFD for the Admin module provides a detailed representation of the administrative functions performed in the MediSmart system. After successful login, the Admin can access and manage different parts of the system. The administrative process is divided into seven major sub-processes: User Management, Medicine Management,

Category Management, Stock Management, Prescription Review & Approval, Order Management, and Payment Reports.

The User Management process allows the Admin to view and manage registered user information. Medicine Management is used to add, update, delete, and maintain medicine details, while Category Management handles the different medicine categories available in the system. Stock Management helps the Admin monitor medicine quantities and update stock information when medicines are added or removed.

The Prescription Review & Approval process allows the Admin to review prescriptions submitted by users and manage their approval status. Order Management enables the Admin to view and update customer orders and their status. The Payment Reports process provides access to payment-related information and reports, helping the Admin monitor completed and pending transactions.

Each administrative process communicates with its corresponding data store, including Users, Medicines, Categories, Stock, Prescriptions, Orders, and Payments. The Admin can read, update, and manage the required information through these processes.

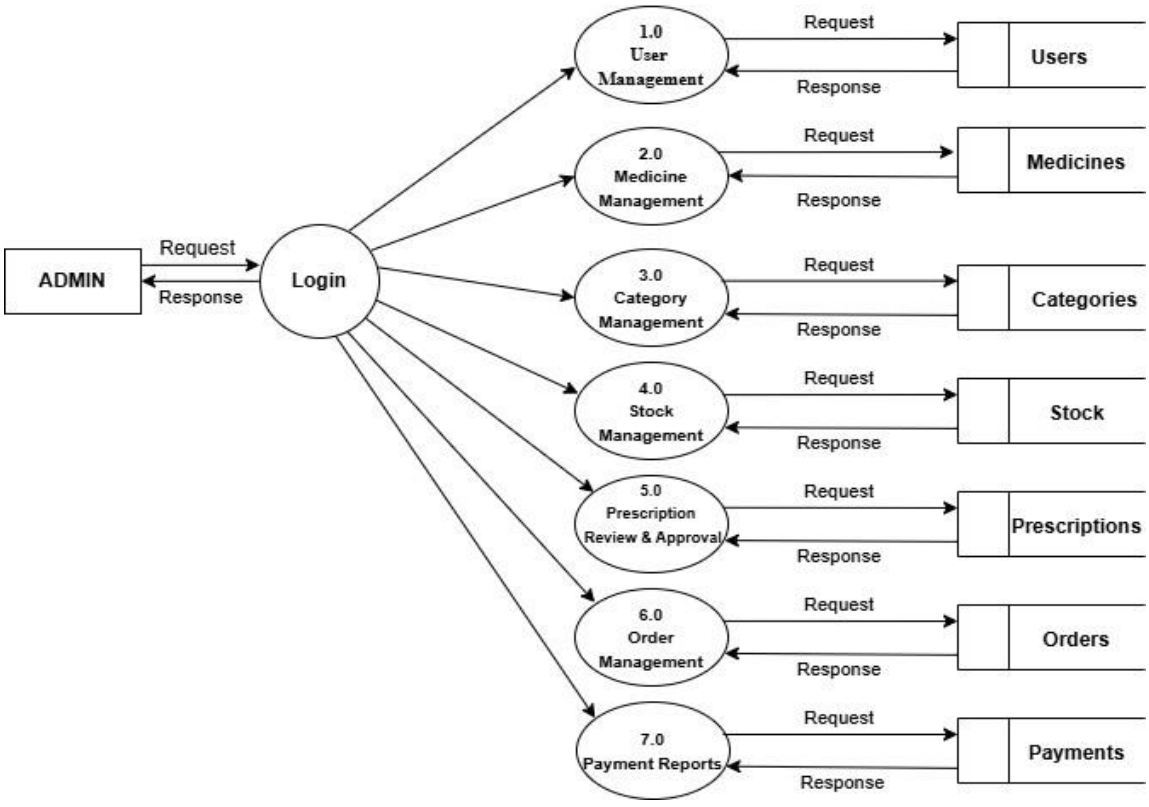


Figure 4.1.3: Level 1 DFD: Admin

4.2 Use Case Diagram

The use case diagram of MediSmart shows two actors, User and Admin. The User can register, search medicines, upload prescriptions, place orders, make payments, and track orders. The Admin manages users, medicines, stock, prescriptions, orders, and payment reports. Prescription handling is shared, with the User uploading the prescription and the Admin reviewing and approving or rejecting it.

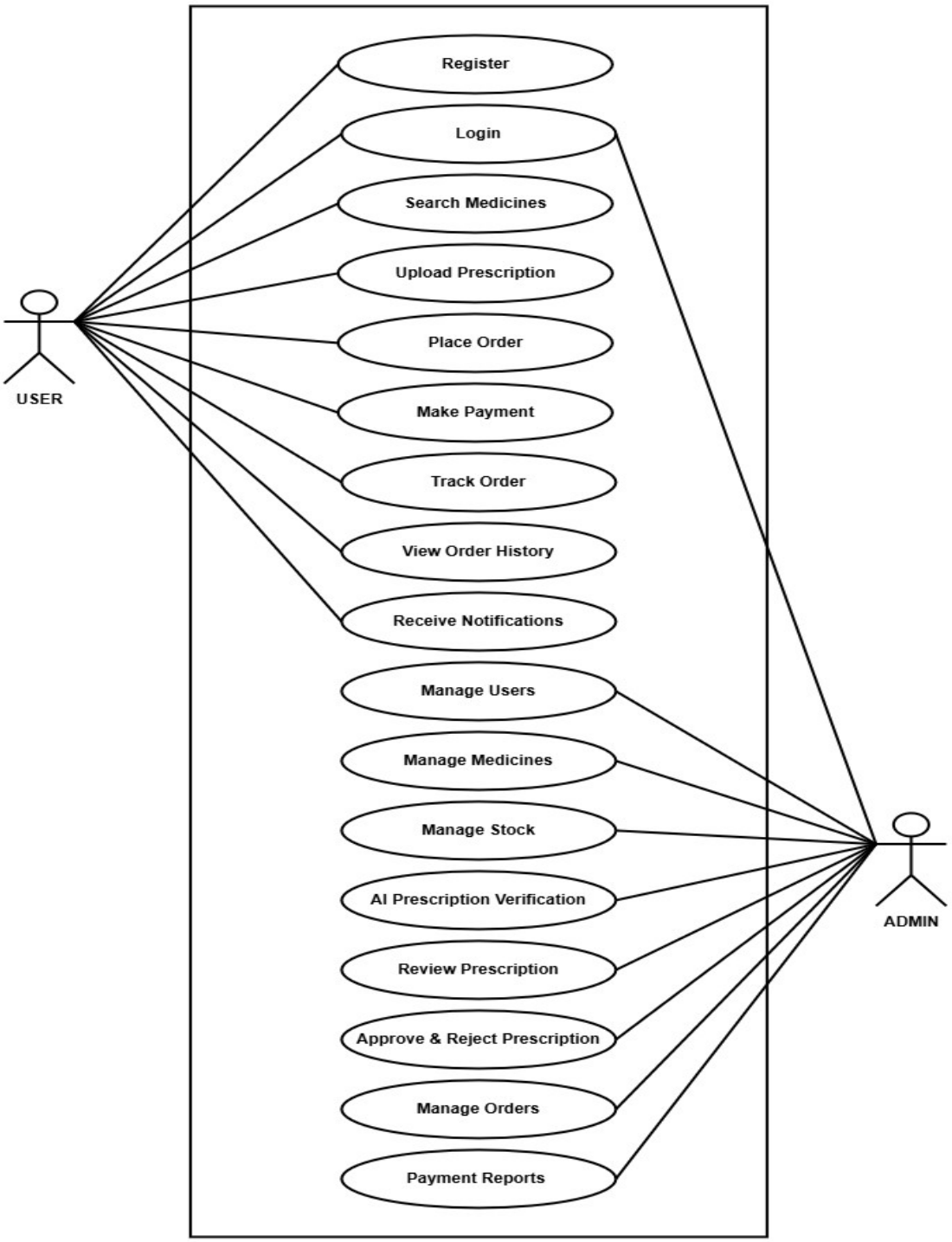


Figure 4.2: Use case diagram

4.3 Class Diagram

Class Diagram represents the main classes of the MediSmart system, including User, Admin, Medicine, Category, Prescription, Order, and Payment, along with their attributes, operations, and relationships. The User class handles registration, login, medicine search, prescription upload, order placement, payment, and order tracking. The Prescription class stores prescription details and manages verification status, while the Medicine and Category classes maintain the pharmacy catalogue. The Order class manages order creation and tracking, and the Payment class records payment details and status. The Admin class provides functions for managing users, medicines, stock, prescriptions, orders, and reports. The relationships between these classes show how the different components of MediSmart are connected and provide a clear structural view of the system.

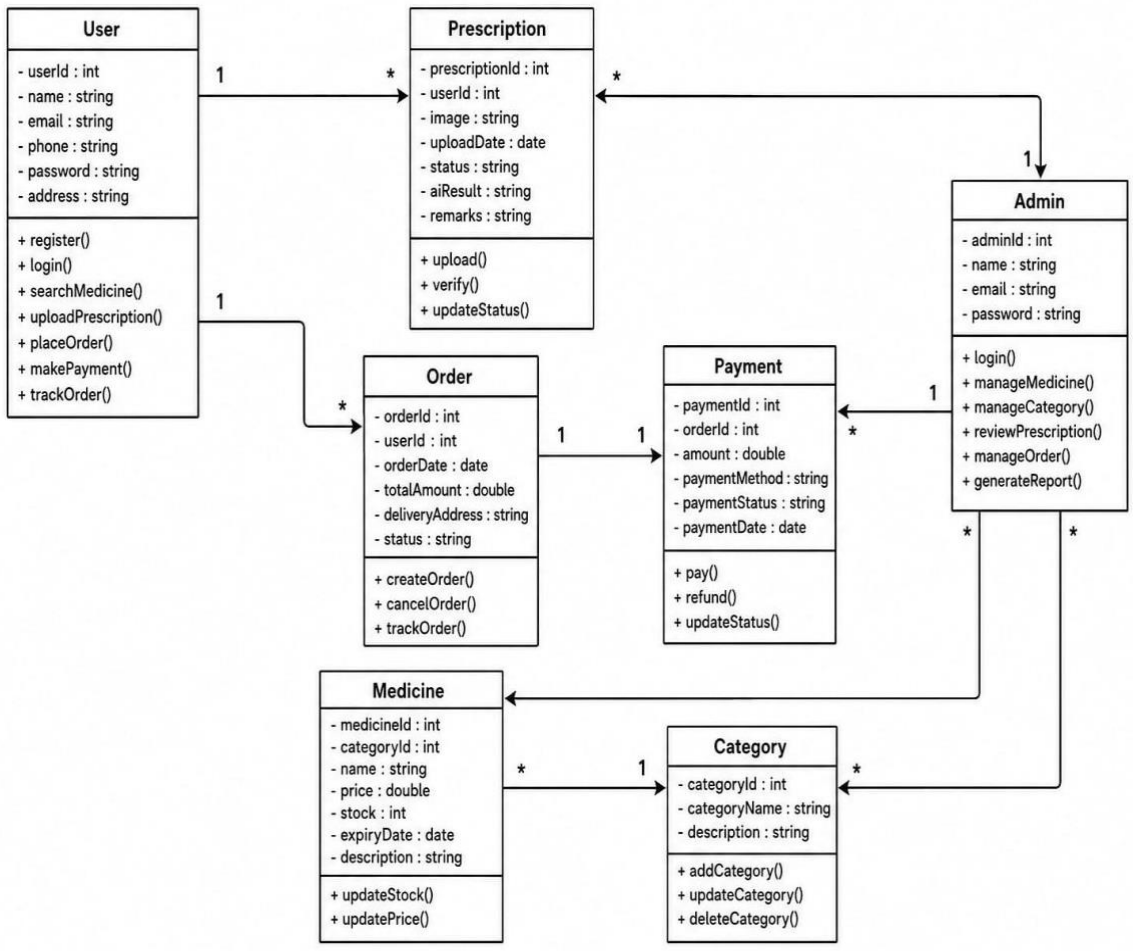


Figure 4.3: Class diagram

4.4 State Diagram

The State Diagram of MediSmart represents the different states and transitions followed by the User and Admin while interacting with the system. On the User side, the process starts with Register/Login and continues through Search Medicines, View Medicine Details, Upload Prescription, AI Prescription Verification, Place Order, Make Payment, Order Confirmation, Order Tracking, and View Order History. If the prescription is rejected, the user returns to the upload stage for resubmission, while an approved prescription allows the process to continue. On the Admin side, the flow begins with Admin Login and proceeds through Review Prescription, Approve/Reject Prescription, Manage Medicines, Manage Orders, Generate Reports, and Logout. Thus, the diagram shows the major state transitions and workflow of both User and Admin in MediSmart.



Figure 4.4: State diagram

4.5 Activity Diagram

The Activity Diagram represents the workflow of both the User and Admin in the MediSmart system. The User activities include registration or login, searching medicines, uploading prescriptions, placing orders, making payments, and tracking orders. The Admin activities include managing users, medicines, stock, prescriptions, orders, and payment reports. The diagram shows the sequence of these activities and the decisions involved during prescription verification and order processing, providing a clear overview of the overall system workflow.

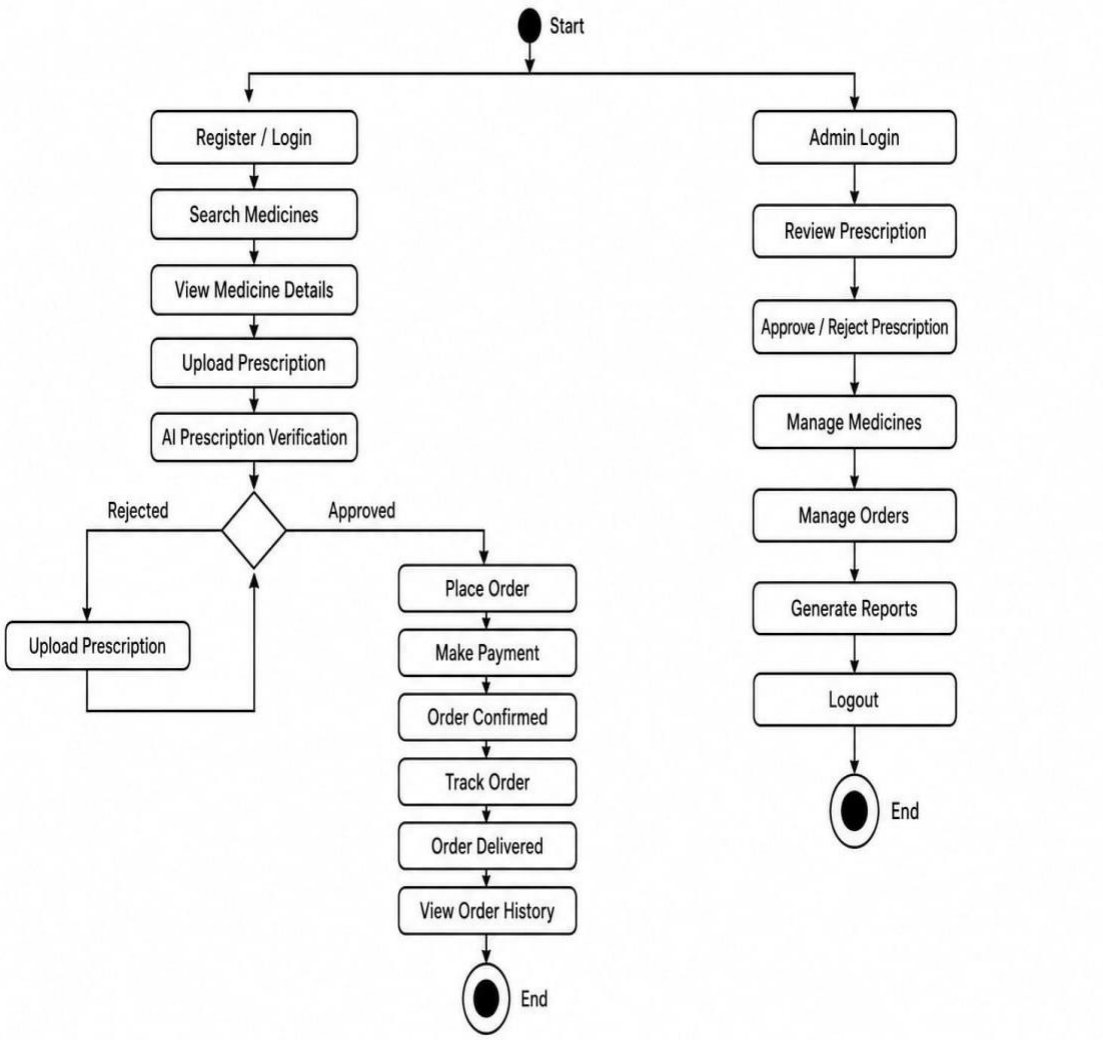


Figure 4.5: Activity diagram

4.6 Sequence Diagram

The Sequence Diagram represents the interaction between the User, System, AI Verification Module, Admin, Database, and Payment Gateway during the order process. The User first registers or logs in, searches for medicines, and uploads a prescription. The system sends the prescription to the AI Verification Module for verification. If the result is doubtful or invalid, it is forwarded to the Admin for review and approval or rejection. Once approved, the User places an order and completes payment through the Payment Gateway. The system stores the required information in the Database and allows the User to track the order and receive status notifications.

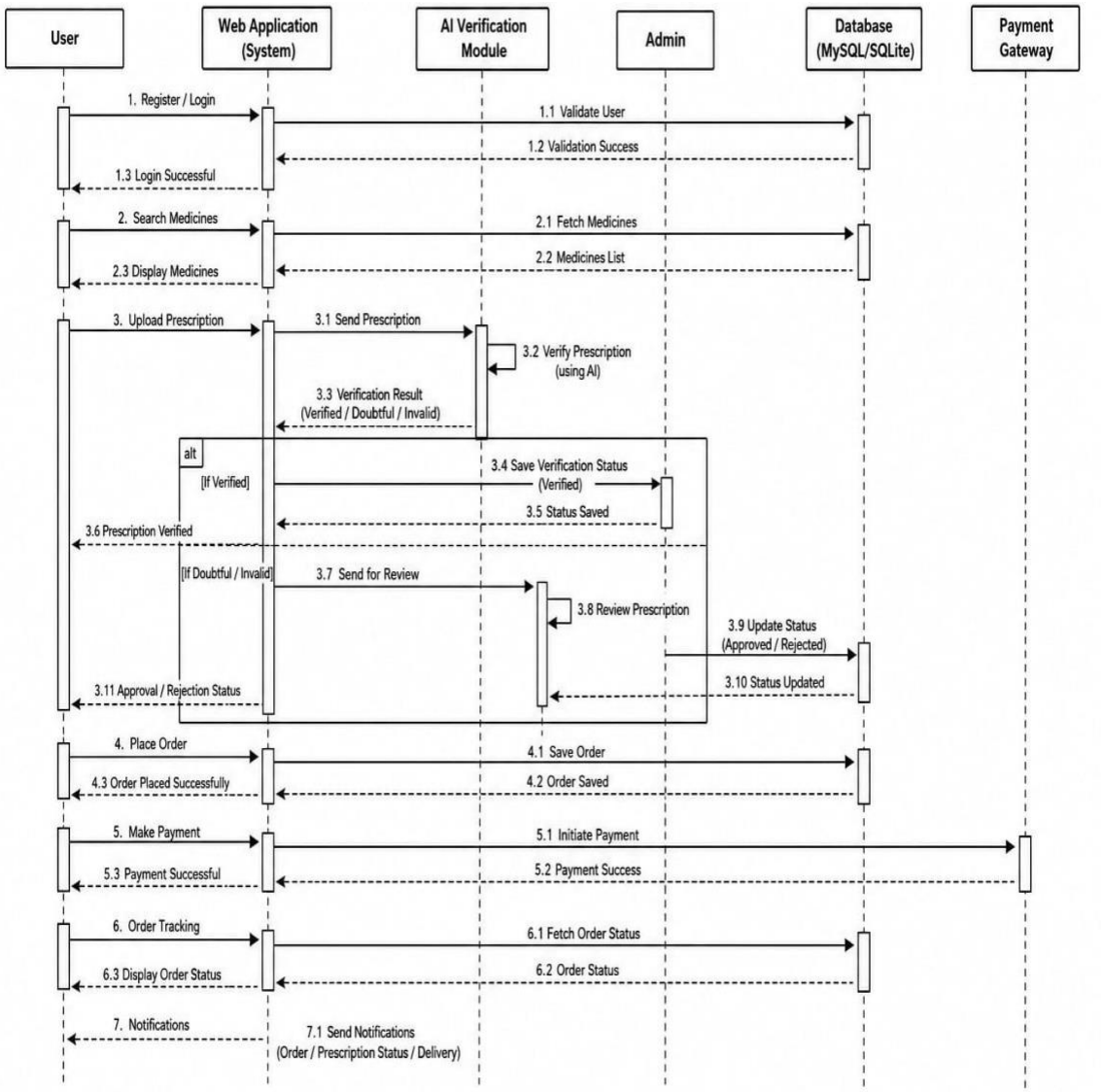


Figure 4.6: Sequence diagram

CHAPTER 5
METHODOLOGY

5.1 Module Descriptions

MediSmart consists of interconnected modules for user management, medicine search, prescription upload and verification, order processing, chatbot assistance, and stock and expiry tracking. The system allows users to manage medicines and prescriptions, place and track orders, while administrators can review prescriptions and manage medicines, stock, orders, and other pharmacy-related activities.

User Module

The User Module is responsible for customer registration, authentication and basic profile management. A new customer can create an account by providing details such as full name, email address, phone number and password. After successful registration, the user can log in using the registered email address and password. Once authenticated, the user is directed to a personalised dashboard that provides access to important functions such as browsing medicines, uploading prescriptions and managing the shopping cart. This module ensures that customer-related information and access to the system are handled in an organised manner.

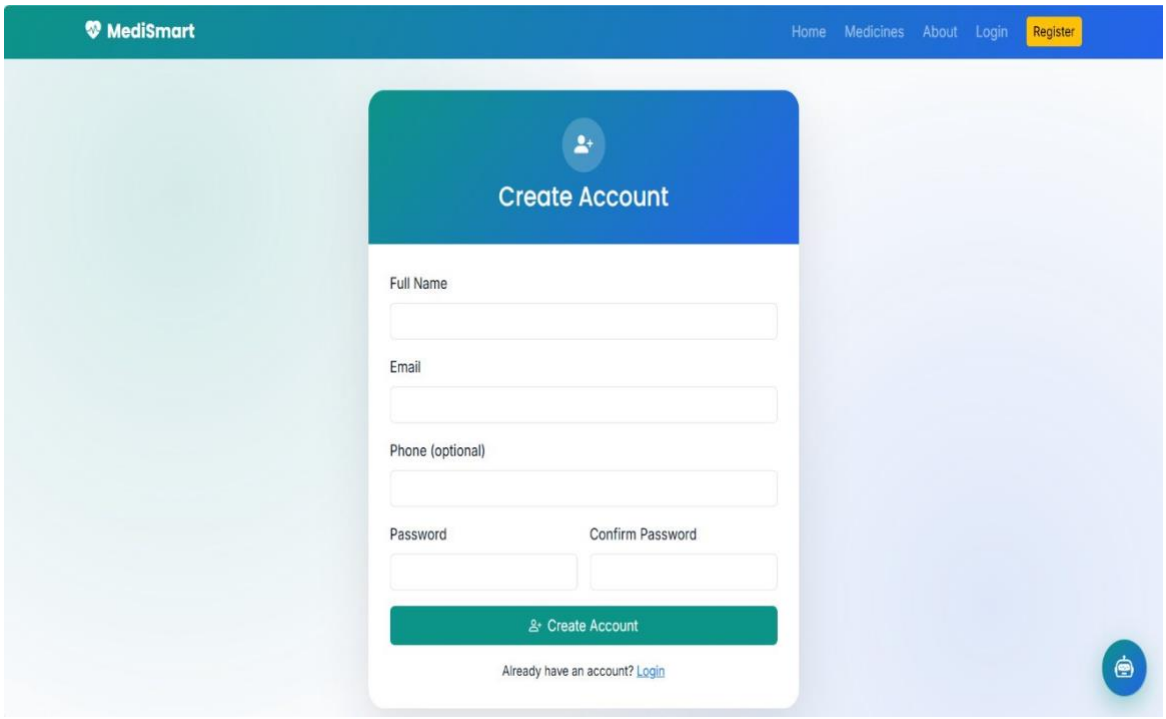


Fig. 5.1.1 User Registration Page

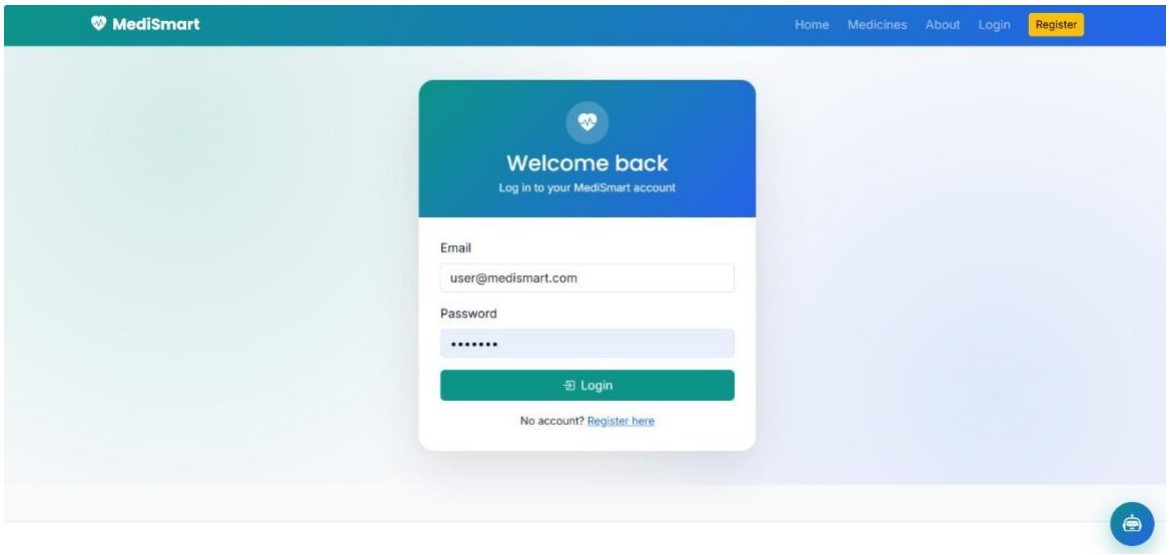


Fig. 5.1.2 User Login Page

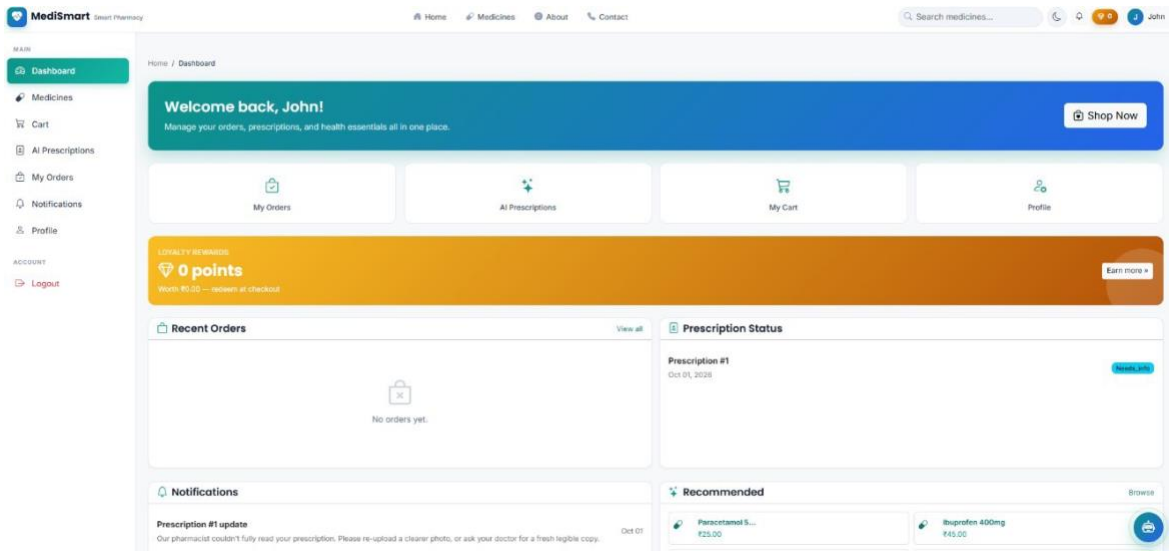


Fig. 5.1.3 User Dashboard

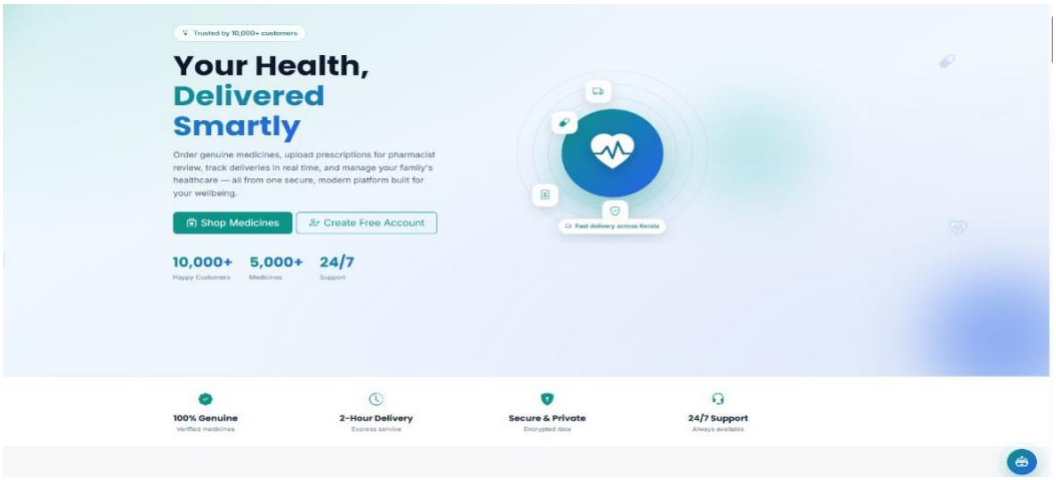


Fig. 5.1.4 Home Page

Prescription Module

The Prescription Module manages the uploading and verification of prescriptions. A user can upload a photograph or scanned copy of a prescription along with an optional note for the pharmacist. The uploaded prescription is processed using the AI Processing Module, which uses OCR and AI techniques to extract and interpret the information contained in the prescription. The system produces a verification status based on the AI-assisted analysis. When the result is doubtful or invalid, the prescription is forwarded to the pharmacy administrator for manual review. The administrator can then approve or reject the prescription, and the updated status is communicated to the user.

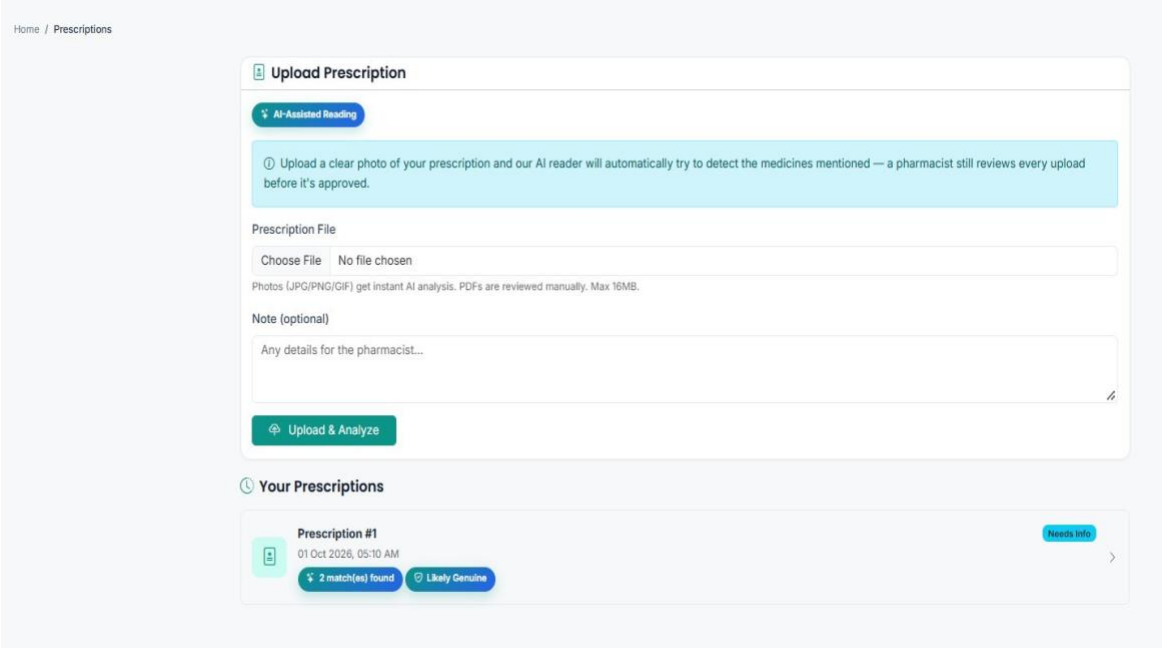


Fig. 5.1.5 Upload Prescription Screen

Order Module

The Order Module provides the functionality required for searching medicines, adding medicines to a cart and placing orders. Users can browse the medicine catalogue and filter medicines according to name or category. When a prescription is required for a particular medicine, the system checks whether the corresponding prescription has been verified or approved before allowing the order to proceed. Users can select either home delivery or take-away collection according to their preference. The module also maintains the order status so that users can track the progress of their orders.

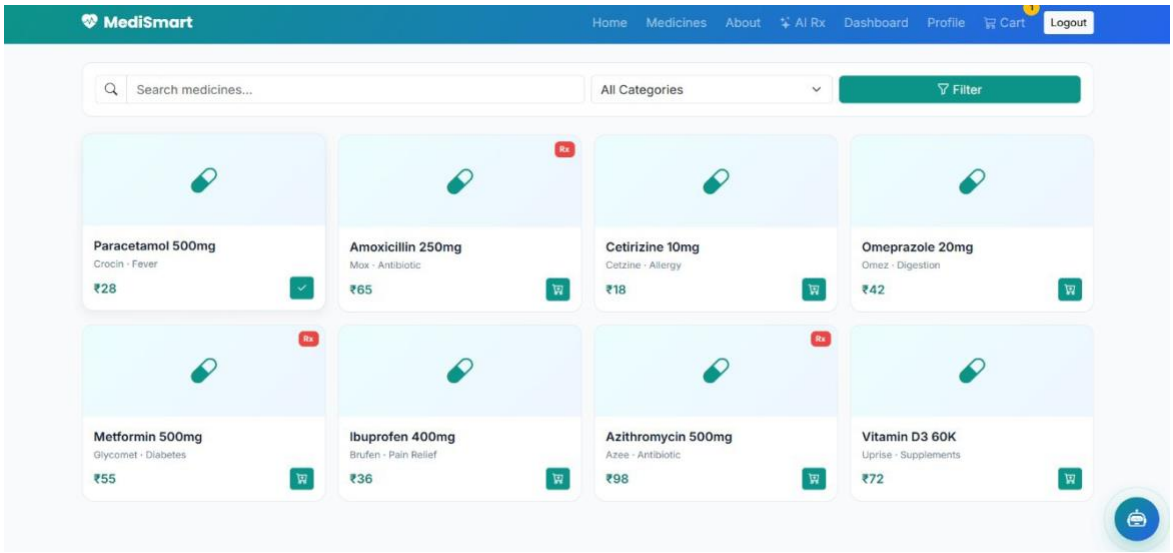


Fig. 5.1.6 Medicine Catalogue and Search

Chatbot Module

The Chatbot Module provides automated assistance for common customer queries. It uses a rule-based approach supported by basic Natural Language Processing techniques such as keyword and intent matching. The chatbot can respond to queries related to medicines, prescriptions and delivery. It also provides quick-reply options for commonly required information. This module reduces the need for customers to contact pharmacy staff directly for routine questions and provides a simple first level of customer support.

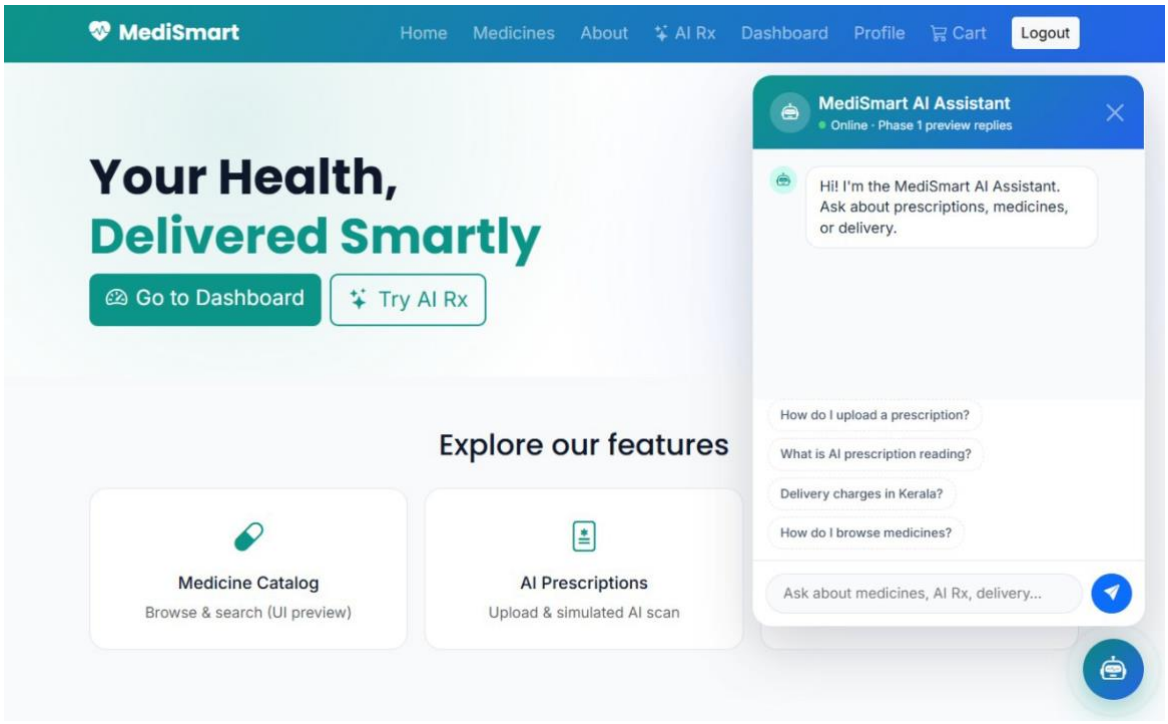


Fig. 5.1.7 AI Assistant (Chatbot)

Stock Module and Expiry Tracking Module

The Stock Module and Expiry Tracking Module are mainly used by the pharmacy administrator to maintain medicine availability and expiry information. Stock quantities and medicine details are stored centrally in the database and can be updated when medicines are received or sold. The expiry tracking function stores the expiry date associated with each medicine and helps the administrator identify medicines approaching their expiry date. Such medicines can be prioritised for sale or removed from the active catalogue when necessary. This module addresses the problems associated with manual stock recording and late identification of expired medicines.

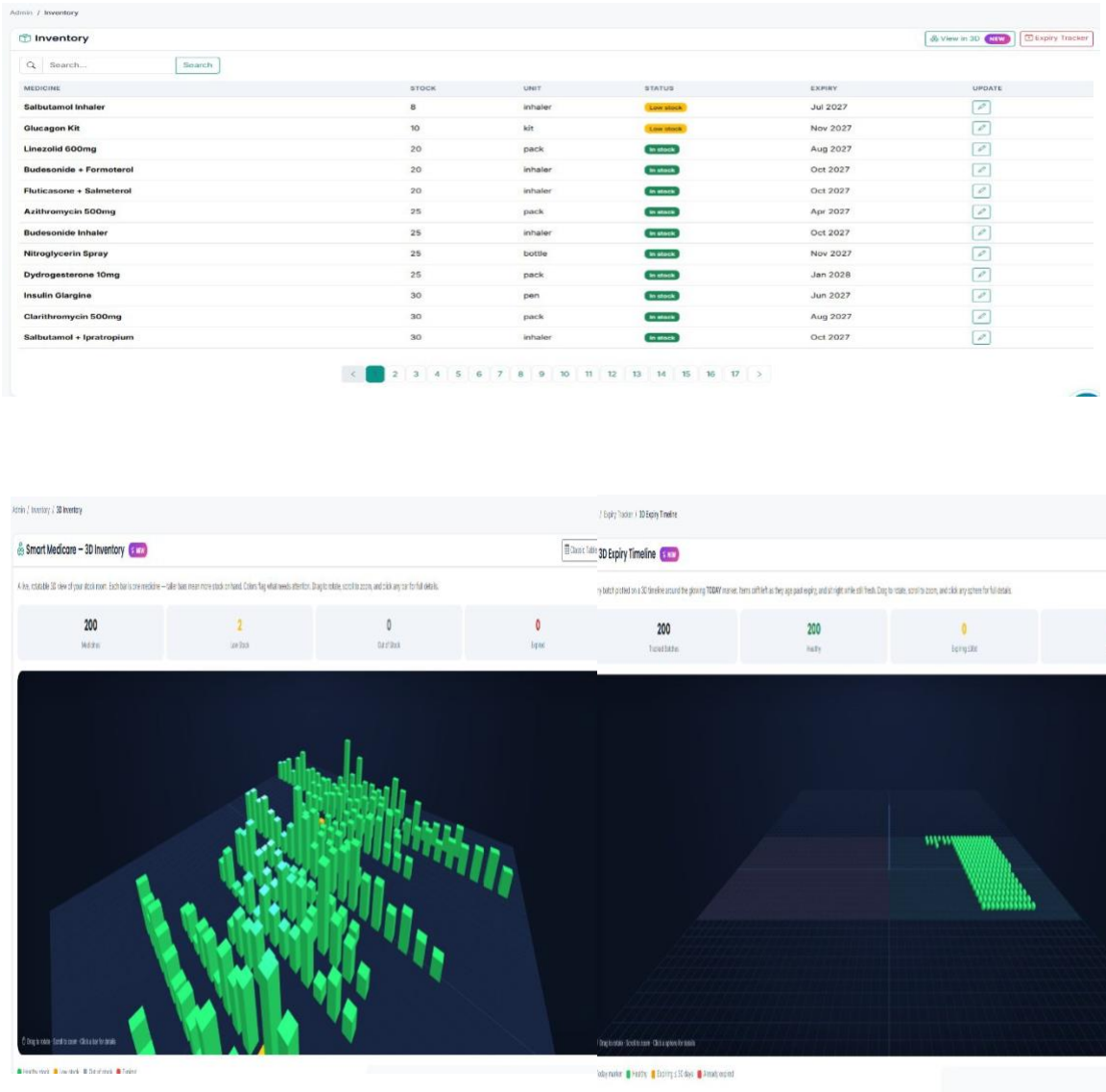


Fig. 5.1.8 Stock manage & 3D Expiry Timeline

5.2 Algorithms

The AI Processing Module is an important component of MediSmart and is used mainly for prescription reading and verification. The system uses Tesseract OCR, TrOCR and the Google Gemini API as part of the prescription verification process. Each technology performs a different role in the processing pipeline. In addition, a rule-based chatbot supported by basic NLP techniques is used to provide automated assistance to users.

5.2.1 Tesseract OCR Algorithm

To extract text from the uploaded prescription image.

Steps:

1. The user uploads a prescription image through the Prescription Module.
2. The system receives and preprocesses the uploaded image.
3. Tesseract OCR analyses the image and identifies the characters present in it.
4. The recognised characters are converted into machine-readable text.
5. The extracted text may include medicine names, dosage instructions and other prescription details.
6. The extracted text is passed to the next stage of the prescription verification process.

5.2.2 TrOCR Algorithm

To improve text recognition from prescription images, especially when the image contains handwriting or quality issues.

Steps:

1. The uploaded prescription image is provided as input to the TrOCR model.
2. The image is processed by the transformer-based OCR model.
3. The encoder extracts relevant visual features from the prescription image.
4. The decoder processes these features to recognise the text.
5. The recognised text is generated as machine-readable output.
6. The extracted information is used along with the OCR results for further prescription interpretation.

5.2.3 Google Gemini API Algorithm

To interpret the extracted prescription information and assist in prescription classification. It also helps identify unclear or inconsistent prescription details to support the verification process.

Steps:

1. The OCR stage produces the text extracted from the prescription.
2. The extracted text and relevant prescription information are provided to the Google Gemini API.
3. Gemini analyses the provided information.
4. The system identifies medicine names and relevant prescription details.
5. The prescription is classified into one of three categories: Verified, Doubtful or Invalid.
6. If the prescription is Verified, it can proceed to the ordering workflow.
7. If the prescription is Doubtful or Invalid, it is forwarded to the administrator for manual review.
8. The final verification status is stored and communicated to the user.
9. The system checks the extracted text for missing or unclear prescription details.
10. The identified medicine names and prescription information are compared with the available validation rules.
11. The administrator reviews doubtful prescriptions and makes the necessary verification decision.
12. The system updates the prescription status in the database and allows the user to view the final result.
13. The system maintains the prescription verification history for future reference.
14. The verification results help the system determine whether the user can continue with the medicine-ordering process.

5.3 Architecture

MediSmart follows a layered client-server architecture that separates the presentation, application logic, AI processing and data storage responsibilities. The main technologies used in the system are HTML, CSS, JavaScript and Bootstrap for the frontend, Python with Flask for the backend, SQLite for data storage, and Tesseract OCR, TrOCR and the Google Gemini API for AI-assisted prescription processing. This separation makes the system

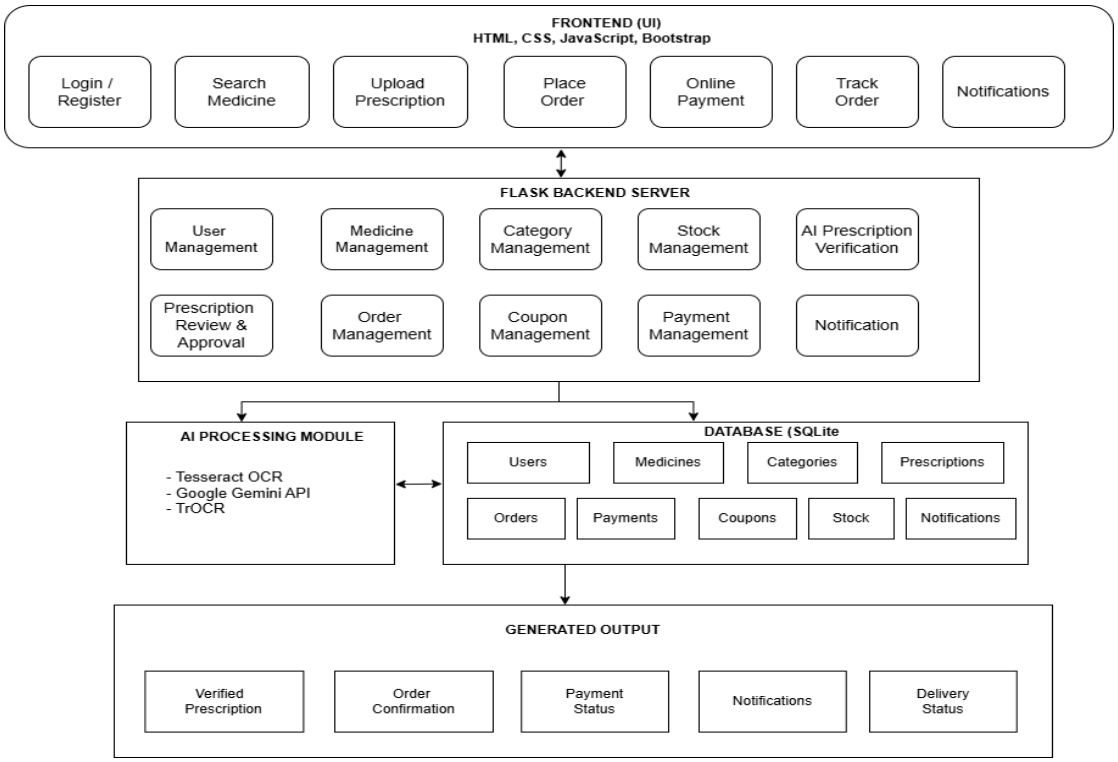


Figure 5.3.1: System architecture

Presentation Layer

The Presentation Layer provides the interface through which customers and administrators interact with the MediSmart system. HTML is used to structure the web pages, while CSS and Bootstrap are used to provide styling and responsive layouts. JavaScript is used for client-side interaction and functions such as form validation and chatbot interaction. This layer mainly captures user input and displays the responses received from the application layer.

Application Layer

The Application Layer is implemented using the Flask framework and contains the main business logic of the MediSmart system. It manages functions such as user management, medicine management, category management, stock management, prescription verification, prescription review, order management, coupon management, payment management and notifications. Keeping these responsibilities within the application layer allows different functions to be developed and maintained independently.

AI Processing Layer

The AI Processing Layer is responsible for prescription reading and verification. It contains the Tesseract OCR, TrOCR and Google Gemini API components used in the prescription processing workflow. This layer is separated from the main application logic because prescription processing is a distinct and safety-sensitive operation. The separation also allows the AI components to be tested, modified or upgraded independently without affecting the main order and pharmacy management functions.

Data layer

The Data Layer uses an SQLite database to store the information required by the MediSmart system. The database contains records related to users, medicines, categories, prescriptions, orders and payments, along with stock, coupon and notification information. SQLite was selected because it does not require a separate database server and is suitable for the scale of the mini project. The database structure follows the relationships defined in the Entity-Relationship diagram.

5.4 Case Study

MediSmart is developed through requirement analysis, system design, front-end and back-end development, AI-assisted prescription verification, and testing. The front end uses HTML, CSS, JavaScript, and Bootstrap, while Python Flask is used for the back end and SQLite for data storage.The workflow starts with user registration and login. Users can search medicines, check availability, and upload prescriptions when required. OCR extracts the prescription text, and the Google Gemini API analyses the information to assist in classifying prescriptions as Verified, Doubtful, or Invalid. Prescriptions requiring further verification are forwarded to the administrator for manual review.

After approval, users can place orders and select delivery or take-away options. The administrator manages prescriptions, medicines, stock, expiry details, and orders. The system also allows users to track orders and view their order history. Overall, MediSmart aims to simplify pharmacy operations, reduce manual work, and provide a convenient medicine-ordering experience.

CHAPTER 6
IMPLEMENTATION DETAILS

6.1 Implementation Procedures

MediSmart was implemented in a structured and iterative manner, beginning with requirement analysis and system design, followed by front-end development, back-end and database implementation, AI integration, system integration and testing. The development was carried out module by module so that each component could be reviewed and refined before being connected with the other components. This approach helped in tracking the development progress of individual pages and modules and ensured that the implemented features remained consistent with the requirements and system design described in the earlier chapters.

The front end of MediSmart was developed using HTML, CSS, JavaScript and Bootstrap. The main user interfaces, including the Login, Registration, Home, Medicine Search, Prescription Upload, User Dashboard and Chatbot pages, were developed and progressively refined. Responsive design techniques were used so that the application could be viewed on different screen sizes. Bootstrap components and grid layouts were used to maintain a consistent interface throughout the application. T

he developed interfaces provide the basic navigation and interaction required for users to register, log in, search medicines, upload prescriptions and access other available features. The back-end implementation was developed using Python with the Flask framework. The database was designed based on the ER diagram and implemented using SQLite.

The database stores important records such as users, medicines, categories, prescriptions, orders and payments. The Flask application was organised into separate functional areas for user management, medicine management, category management, stock management, prescription verification, order management, coupon management, payment management and notifications. This modular structure allows the different business functions to be developed and tested separately before being integrated into the complete application.

The AI-assisted prescription verification feature was designed by connecting the prescription upload interface with the AI Processing Module.

The module uses Tesseract OCR, TrOCR and the Google Gemini API as described in the methodology chapter. The uploaded prescription is intended to pass through the OCR and AI processing stages so that the prescription information can be extracted and verified. At the stage covered by this report, the AI verification feature was exercised using simulated or preview-level responses at the interface level. Full end-to-end integration with the live OCR and Gemini API services remains part of the planned further development.
The individual modules were then integrated to form the overall MediSmart workflow.

The User Module was connected with the Order Module so that authenticated users could proceed with ordering. The Order Module was connected with the Prescription Module so that prescription-required medicines could be associated with a verified or administrator-approved prescription. The system architecture also provides for payment and order tracking functions as part of the complete order lifecycle. After integration, the implemented interfaces and available workflows were tested to identify functional issues and to verify whether the system behaved according to the expected requirements. This implementation process resulted in a working prototype with the major customer-facing interfaces developed and an overall front-end completion of approximately 80%.

6.1.1 Development Approach

The implementation of MediSmart was carried out in a structured and modular manner. The development process followed the stages identified in the system methodology, including requirement analysis, system design, front-end development, back-end development, AI processing and testing. Each stage was considered separately before integrating the different components of the system. The main objective of the implementation was to develop a web-based pharmacy management system that could provide a convenient platform for users to search medicines, upload prescriptions, place orders and receive assistance through the chatbot. At the same time, the system was designed to support pharmacy-side activities such as prescription review, stock management and expiry tracking.

The front-end interface was developed using HTML, CSS, JavaScript and Bootstrap. These technologies were used to create responsive and user-friendly pages. The back-end functionality was designed using Python with the Flask framework. SQLite was selected as the database for storing user, medicine, prescription, order, payment, stock and other related

information.The development was carried out incrementally. Core functions such as registration, login, medicine search and catalogue access were implemented first. Other functions such as prescription upload, chatbot support and dashboard features were developed as additional modules. This modular approach made it easier to identify incomplete areas and test individual functions separately

6.1.2 Front-End Implementation

The front-end of MediSmart provides the main interaction between the user and the system. The interface was designed to make common pharmacy-related activities simple and accessible. HTML was used to structure the web pages, while CSS and Bootstrap were used for layout, styling and responsive design. JavaScript was used to provide interactive behaviour wherever required. The registration interface allows a new user to create an account by entering details such as name, email address, phone number and password. The login interface allows registered users to access the system using their credentials. After successful login, the user can access the dashboard and other available functions.

The home page provides access to the major services of MediSmart. Users can navigate to the medicine catalogue, prescription section, cart and other available services. The medicine search interface allows users to find medicines and filter them based on categories. The prescription upload interface allows users to select and upload a prescription image and provide an optional note. The dashboard provides information related to the user's account and available pharmacy services. The chatbot interface provides a simple conversation-based method for obtaining answers to common queries.

The front-end was developed with usability in mind so that users can move between different functions without unnecessary complexity. The interfaces were also tested individually to identify navigation, input and display-related issues.

6.1.3 Back-End and Database Implementation

The back-end of MediSmart is based on Python and the Flask framework. Flask provides the application structure required to process user requests, handle application logic and communicate with the database. The back-end acts as an intermediate layer between the user interface and the stored data.

The system maintains information related to users, medicines, categories, prescriptions, orders, payments, stock, coupons and notifications. SQLite is used as the database because it is lightweight and suitable for the scale of the mini-project.

The database structure follows the relationships defined during the system design stage. User information is stored separately from medicine and prescription information. Medicines are associated with categories, while orders contain information about the products selected by users.

Prescription records are maintained so that their status can be reviewed and updated. The back-end also supports the main workflow of the application. User authentication is handled through the application logic, while medicine search and category filtering retrieve relevant information from the database. Order-related operations are designed to connect the selected medicines with the corresponding user and order information.

The modular back-end structure makes it possible to maintain different functions independently. It also provides a foundation for future integration of additional services and more advanced database systems.

6.2 Testing Methods and Results

Testing of MediSmart was carried out mainly through functional testing, usability testing and integration-path testing. Functional testing was used to verify whether the implemented screens and features performed their intended operations. Usability testing was used to check whether users could understand and navigate the interfaces for tasks such as registration, login, medicine search and prescription upload. Integration-path testing was used to examine the flow of requests and responses between the user interface and the corresponding system components. The testing was performed based on the level of implementation available at the time of preparing the report and therefore does not represent testing of a fully deployed production system.

The registration and login functions were tested successfully. The registration page accepted the required user information and allowed a new user account to be created, while the login function correctly handled the authentication flow. Medicine search and category filtering were also tested and produced the expected results, allowing users to locate medicines through the available catalogue. These results confirm that the basic customer

navigation and catalogue-related functions were functioning correctly at the tested stage of development.

The prescription upload interface was tested at the user-interface level and was found to behave as expected for uploading and displaying prescription information. However, the complete AI-assisted verification process was not independently verified using a fully connected live Tesseract OCR, TrOCR and Google Gemini API pipeline. The verification response available during testing was at a simulated or preview level. Therefore, the actual accuracy of prescription recognition could not be measured from the current testing results.

The user dashboard and chatbot interfaces were also tested at the interface level. The dashboard interface was available for displaying user-related information, while the chatbot provided preview or quick-reply responses. The chatbot was not tested as a fully dynamic AI/NLP-based service because the complete backend integration was not available at the stage covered by the report. Similarly, administrator-side functions and live payment gateway integration were not independently verified because the required implementation material was not available for testing.

Overall, the testing results show that the core registration, login, medicine search and filtering functions were successfully verified. Prescription upload, dashboard and chatbot features were verified at the interface level, while their complete backend functionality requires further testing. The AI prescription verification, live payment processing and administrator functions therefore remain to be fully verified.
These results are consistent with the development progress recorded in the report, where Login and Registration reached full front-end completion, while Prescription Upload, Dashboard and Chatbot were at lower completion levels. The testing therefore confirms the working foundation of MediSmart while also identifying the components that require further integration and end-to-end validation.

6.1.1 Testing Strategy

Testing was carried out to check whether the major functions of MediSmart behave according to the expected workflow. The testing focused mainly on the available user interface and functional behaviour of the implemented modules. The main objective was to identify errors, verify the working of important features, and ensure that the system provides a smooth user experience.

Registration, login, medicine search, and category filtering were tested using valid inputs and expected outputs. Prescription upload, prescription history, and chatbot functions were also checked at the interface level. The navigation between different pages and the display of medicine details were examined to ensure that users could access the required features easily. The administrator interface was also reviewed to check the accessibility of prescription management, medicine management, and order management functions.

The testing process helped identify functions that were ready for verification and functions that still required additional back-end or external-service integration. Any observed issues were noted for further improvement. Overall, the testing strategy helped evaluate the current implementation of MediSmart and identify areas that require further development to improve functionality, reliability, and usability.

6.2.1 Front-End Testing

The front-end components were tested individually to ensure that users could navigate between the available pages and interact with the provided controls. Input fields were checked using appropriate values, and navigation elements were examined for expected behaviour.

Registration and login achieved complete front-end implementation in the current project stage. The home page, medicine search, prescription upload, dashboard and chatbot were implemented to different levels of completion.

The testing results indicate that the core customer-facing navigation is available, while some AI-dependent and backend-dependent features require additional integration and testing before they can be considered fully verified.

6.2.2 Functional Testing

Functional testing was performed using specific test cases to verify whether the implemented features of MediSmart worked according to the expected requirements. Registration with valid information successfully created the required user flow, and login using registered credentials was also successful. These tests helped confirm the basic functionality of user account management.Medicine search and category filtering were tested to ensure that users could locate medicines from the catalogue. These functions produced the expected results. Prescription upload was also checked at the interface level to verify that users could access the upload workflow. However, complete end-to-end AI verification was not independently confirmed at this stage.

Prescription history was checked using the available interface and sample data. The chatbot interface and quick-reply functionality were tested at the available preview level. Direct access to a fully dynamic chatbot response system was not treated as fully verified. The testing results helped identify the completed functions and the features requiring further integration and improvement.

6.2.3 Testing Results

The testing results show that the basic user-facing functions of MediSmart are working at the current development stage. Registration, login, medicine search, and category filtering were successfully verified. These functions provide the foundation for users to access the system, browse the medicine catalogue, and find the required medicines.

Other features, particularly those dependent on AI processing, external APIs, or incomplete backend integration, were verified only at the interface level. Prescription upload, prescription history, and chatbot functions were checked based on their available implementation. Successful display of an interface does not by itself prove complete end-to-end functionality, so these features require further testing and validation.

The testing results therefore provide a clear indication of the current implementation status of the project. They help identify the completed functions and the modules that require further development and integration. Additional testing of prescription verification, database operations, order processing, and external service connectivity would help improve the overall reliability of MediSmart before deployment in a real pharmacy environment.

CHAPTER 7
ADVANTAGES AND DISADVANTAGES

7.1 Advantages

The MediSmart – Smart Pharmacy & Prescription System provides several benefits to customers and pharmacy staff by combining online medicine services, prescription verification, inventory management and customer support in a single platform. The system improves convenience, reduces manual effort and helps pharmacies manage their daily activities more efficiently.

- **Easy Access to Pharmacy Services:** Customers can access pharmacy services through a single online platform without the need to visit the pharmacy for basic enquiries.
- **Online Medicine Availability:** Users can search for medicines and check their availability online, saving time and effort.
- **AI-Assisted Prescription Verification:** Tesseract OCR, TrOCR and Google Gemini API assist in reading and verifying uploaded prescriptions, reducing the initial manual effort.
- **Centralised Stock Management:** The system provides centralised management of medicine stock and helps pharmacy staff monitor available medicines efficiently.
- **Expiry Tracking:** The system helps identify medicines that are approaching their expiry date, which can reduce medicine wastage.
- **Online Medicine Ordering:** Customers can search for medicines and place orders online, providing a convenient purchasing experience.
- **Secure Payment and Order Tracking:** Online payment and order tracking features provide customers with a more transparent and organised ordering process.
- **In-Application Chatbot:** The chatbot provides automated responses to common customer queries and reduces the workload of pharmacy staff.
- **Modular System Architecture:** The separation of frontend, backend, AI processing and database layers makes the system easier to maintain, test and extend.
- **Cost-Effective Technology:** The use of Flask, SQLite, Bootstrap and open-source OCR tools helps keep development costs relatively low and makes the system suitable for small and medium-sized pharmacies.

- **Reduced Paperwork:** Digital storage of customer details, prescriptions and order information helps reduce the use of manual records and paperwork.
- **Faster Order Processing:** The online ordering process helps pharmacy staff receive and manage customer orders in an organised manner.
- **Improved Customer Communication:** The chatbot and online platform provide customers with an additional way to communicate with the pharmacy.
- **Support for Digital Pharmacy Services:** The system encourages the use of digital technology in traditional pharmacy operations and customer services.

7.2 Disadvantages

Although the MediSmart system provides several useful features, it also has certain limitations that need to be considered. Some limitations are related to AI-based prescription processing, database scalability, internet dependency and the need for professional supervision

- **Dependence on Image Quality:** The accuracy of prescription verification depends on the quality of the uploaded prescription image.
- **Difficulty with Handwritten Prescriptions:** Poor handwriting, unclear text or complex prescriptions may reduce the accuracy of OCR-based text extraction.
- **Need for Further Integration:** Some modules, including prescription verification, prescription history and chatbot functionality, require further backend integration and end-to-end testing.
- **Database Scalability:** SQLite may not be suitable for handling very large numbers of simultaneous users and may require migration to a more scalable database in the future.
- **Internet Dependency:** The system requires an internet connection for customers and administrators and does not currently provide a complete offline mode.
- **AI Cannot Replace Pharmacists:** AI-based prescription verification is only a supporting feature and cannot replace the professional judgement of a qualified pharmacist.
- **Administrator Verification Required:** Doubtful or unclear prescriptions require proper review and verification by authorised pharmacy staff.

- **Further Testing Required:** Additional real-world testing and complete end-to-end integration are required before the system can be considered for full-scale production deployment.

- **Dependence on External AI Services:** Some AI-based functions depend on external services, and interruptions or delays may affect their operation.
- **Possibility of Incorrect Extraction:** OCR technology may incorrectly identify medicine names, quantities or other prescription details, requiring manual checking.
- **Different Prescription Formats:** Prescriptions may differ in layout, language and writing style, making automatic processing more difficult.
- **Data Privacy Concerns:** The system handles sensitive customer and prescription information, so proper data protection and access control are necessary.
- **Maintenance Requirements:** Regular software updates, database maintenance and technical support are required to keep the system functioning properly.
- **Limited Offline Functionality:** Users may not be able to access important online services when the internet connection is unavailable.
- **Need for Staff Training:** Pharmacy employees may require training to understand the system and use its different modules correctly.
- **AI Processing Time:** Processing prescription images through OCR and external AI services may take additional time depending on image quality and service response.

CHAPTER 8
RESULT AND DISCUSSION

8.1 Analysis

The analysis of the MediSmart system was carried out based on the implemented modules, system design, functional requirements, and available interface-level testing. The developed prototype successfully establishes the basic workflow of a smart pharmacy and prescription management system. The system provides a structured platform through which users can access pharmacy-related services digitally.

The system includes important customer-facing functionalities such as user registration, login, medicine search, category-based filtering, prescription upload, dashboard access, and chatbot support. These features help to simplify the process of finding medicines and managing prescription-related activities. The interface design also provides a clear and organized workflow for users.

A major feature of MediSmart is the proposed AI-assisted prescription verification module. The system architecture incorporates OCR and AI technologies such as Tesseract OCR, TrOCR, and Google Gemini API for extracting and analysing prescription information. However, complete end-to-end testing using live OCR and AI services was not carried out during the evaluation. Therefore, the AI verification results cannot be considered as fully validated system results.The analysis also shows that the system has been designed using a structured software architecture supported by database design and different system modelling diagrams. DFD, use case, class, activity/state, sequence, ER, and system architecture diagrams were used to represent the system structure and workflow. These design elements provide a clear understanding of the interaction between users, system modules, and data.

Functional testing was performed for the major available features. User registration, login, medicine search, and category filtering were verified at the available implementation level. Prescription upload and order/history-related features were available mainly through the user interface, while some backend functions could not be completely verified. The chatbot also provides basic responses through the available interface.Overall, the analysis indicates that MediSmart successfully establishes the foundation of a smart pharmacy and prescription management system. The customer-side workflow, system architecture,

database structure, and major interfaces have been developed. Further development is required to complete backend integration, validate AI-based prescription verification with live services, improve chatbot functionality, and perform complete administrator-side testing.

The key observations obtained from the analysis of the system include:

- **User Registration and Login:** The system provides registration and login facilities for secure user access and account management.
- **Medicine Search and Filtering:** Users can search for medicines and filter medicines based on available categories, making the medicine selection process easier.
- **Prescription Upload:** Users can upload prescriptions through the system, providing a digital method for prescription submission and management.
- **AI-Assisted Prescription Verification:** The system architecture includes OCR and AI-based technologies for extracting and verifying prescription information. However, complete end-to-end verification with live AI services was not fully tested.
- **Chatbot Support:** A chatbot interface is included to provide basic assistance and quick responses to users. The current implementation is mainly at the interface level.
- **Order and History Management:** - The system provides interfaces for cart, order-related activities, and viewing previous records, although complete backend functionality was not independently verified.
- **Database and System Design:** The system is supported by database design, ER diagrams, DFDs, use case diagrams, class diagrams, activity/state diagrams, sequence diagrams, and system architecture.

- **Functional Testing:** Core functions such as registration, login, medicine search, and category filtering were tested and found to be functioning at the available implementation level.

CHAPTER 9

CONCLUSION

9.1 Summary

This report has presented MediSmart – Smart Pharmacy & Prescription System, a web-based platform designed to connect customers and pharmacies through a single digital platform. The main aim of the project is to simplify pharmacy operations, improve medicine accessibility, and reduce the difficulties associated with traditional pharmacy management. The system addresses common problems such as manual record-keeping, time-consuming medicine searches, difficulties in handling prescriptions, and inefficient stock monitoring. By introducing a digital platform, MediSmart provides a more organized approach to managing pharmacy-related activities.

The system is developed using six major functional modules: User Management, Prescription Management, Order Management, Chatbot, Stock Management, and Expiry Tracking. Each module performs a specific function and contributes to the overall operation of the system. User Management supports customer registration and login, while Prescription Management handles the uploading and processing of prescriptions. Order Management supports medicine ordering, and the Chatbot module provides assistance for basic customer queries. Stock Management helps maintain medicine availability records, whereas Expiry Tracking supports the identification of medicines approaching their expiry dates. These modules work together to provide a structured pharmacy management environment.

An important feature of MediSmart is its AI Processing Module, which uses Tesseract OCR, TrOCR, and the Google Gemini API to support prescription image processing and text extraction. These technologies help reduce the manual effort involved in reading prescription information. The system uses Flask as its backend framework and SQLite for database management. The modular architecture makes the application easier to maintain, modify, and extend in the future. The technical design and implementation details are discussed in Chapters 4 and 5.

The testing activities described in Chapter 6 evaluated the major functions of the application. The registration, login, medicine search, and medicine catalogue features were tested to verify their basic operation. The testing process helped identify the functional

status of the implemented modules and provided an understanding of the system's performance under the tested conditions. The AI-assisted prescription verification and chatbot features were also examined at the interface level. However, complete end-to-end integration and testing of these features remain necessary to establish their full operational reliability.

The literature review presented in Chapter 3 examined existing research related to online pharmacy platforms, machine-learning-based prescription recognition, handwritten prescription processing, and digital pharmacy services. The review highlighted the importance of combining prescription processing, medicine ordering, customer support, and pharmacy inventory management in a unified system. The research gap identified in Section 3.7 provided the basis for developing MediSmart as a modular platform that brings these functions together. The project therefore applies ideas from existing research to a practical pharmacy management application.

Chapter 8 presented a comparative discussion of the proposed system, the existing manual pharmacy system, and the related research works. The comparison highlighted how digital processing can support more organized medicine management, prescription handling, and customer interaction. It also helped identify the areas in which the current implementation requires further development. In particular, complete AI integration, advanced administrative functions, and additional testing remain important areas for improvement.

Overall, MediSmart has established a foundation for a smart pharmacy and prescription management platform by combining web technologies, database management, and AI-based processing techniques. The project has addressed its main architectural and foundational objectives through the development of its core modules. Although some advanced features require further integration, testing, and refinement, the implemented system demonstrates the potential of using digital technologies to simplify pharmacy-related activities. The future development of MediSmart can focus on improving accuracy, security, scalability, and real-time functionality, thereby making the platform more suitable for practical pharmacy environments.

9.2 Limitations

- The system requires a stable internet connection to access its online features.
- The accuracy of prescription data extraction depends on the quality and clarity of the uploaded prescription images.
- Poor-quality, blurred or handwritten prescriptions may affect the accuracy of OCR processing.
- The system has limited scalability for handling a large number of users simultaneously.
- Real-time stock updates may require further improvement to ensure accurate medicine availability.
- The system does not currently provide complete integration with external hospital management systems.
- Advanced security features and additional data protection mechanisms require further enhancement before large-scale deployment.
- The system has not yet been evaluated through large-scale real-world testing in pharmacies.

9.3 Future Scope

- Advanced AI Prescription Verification: Improve the AI module to accurately read and verify handwritten and printed prescriptions using advanced OCR and AI technologies.
- Dynamic Chatbot Integration: Develop an intelligent chatbot that can provide real-time responses to customer queries related to medicines, orders, and pharmacy services.
- Online Payment Integration: Add secure online payment options to make medicine purchasing easier and faster.
- Delivery Tracking: Integrate delivery services to allow customers to track their medicine orders in real time.
- Mobile Application: Develop Android and iOS applications to make the system more accessible to customers and pharmacy staff.
- Advanced Inventory Management: Implement automatic stock updates, low-stock alerts, and medicine expiry notifications to improve pharmacy management.

- Hospital Integration: Connect the system with hospital management systems to support smoother prescription handling and information exchange.
- Enhanced Security: Introduce advanced authentication, data encryption, and access control to protect sensitive customer and prescription information.
- Multi-Branch Pharmacy Support: Extend the system to support multiple pharmacy branches with centralized stock and order management.
- Performance and Scalability: Upgrade the database and system architecture to support more users, larger amounts of data, and higher transaction volumes.

9.4 Applications

- **Retail Pharmacies:** Helps pharmacies manage medicine details, stock availability, orders, and expiry information.
- **Online Medicine Purchasing:** Allows customers to browse available medicines and place orders through an online platform.
- **Prescription Management:** Helps customers upload prescriptions and supports the process of reading and verifying prescription details.
- **Medicine Stock Management:** Helps pharmacy staff monitor medicine availability and maintain stock information.
- **Expiry Date Tracking:** Helps identify medicines nearing their expiry dates and supports better inventory management.
- **Customer Support:** Provides a chatbot interface to assist customers with basic queries about the system and pharmacy services.
- **Order Management:** Helps pharmacies view and manage customer orders more systematically.

CHAPTER 10

APPENDICES

10.1 Sample Code

routes.py

```python
app > auth > routes.py > register
from app.validation import (
    validate_name, validate_email, validate_signup_email, validate_phone, va
)

bp = Blueprint("auth", __name__, url_prefix="/auth")


@bp.route("/login", methods=["GET", "POST"])
def login():
    if current_user.is_authenticated:
        return redirect(_home_for(current_user))
    form = {"email": ""}
    if request.method == "POST":
        email, email_err = validate_email(request.form.get("email", ""))
        password = request.form.get("password", "")
        remember = bool(request.form.get("remember"))
        form["email"] = request.form.get("email", "")
        if email_err:
            flash(email_err, "danger")
            return render_template("auth/login.html", form=form)
        if not password:
            flash("Password is required.", "danger")
            return render_template("auth/login.html", form=form)
        user = User.query.filter_by(email=email).first()
        if user and user.check_password(password):
            if not user.is_active_user:
                flash("This account has been disabled. Please contact support
                return render_template("auth/login.html", form=form)
            login_user(user, remember=remember)
            flash("Welcome back, %s!" % user.name, "success")
            return redirect(_home_for(user))
        flash("Invalid email or password.", "danger")
    return render_template("auth/login.html", form=form)


@bp.route("/register", methods=["GET", "POST"])
```

routes.py

```python
app > auth > routes.py > register


@bp.route("/register", methods=["GET", "POST"])
def register():
    if current_user.is_authenticated:
        return redirect(_home_for(current_user))
    form = request.form if request.method == "POST" else {}
    if request.method == "POST":
        name, name_err = validate_name(request.form.get("name", ""))
        email, email_err = validate_signup_email(request.form.get("email", "
        phone, phone_err = validate_phone(request.form.get("phone", ""), requ
        address, address_err = validate_address(request.form.get("address", "
        password, password_err = validate_password(
            request.form.get("password", ""),
            confirm=request.form.get("confirm", ""),
        )
        errors = [e for e in (name_err, email_err, phone_err, address_err, pa
        if errors:
            for e in errors:
                flash(e, "danger")
            return render_template("auth/register.html", form=request.form)
        if User.query.filter_by(email=email).first():
            flash("An account with that email already exists.", "danger")
            return render_template("auth/register.html", form=request.form)
        user = User(name=name, email=email, phone=phone or None, address=addr
        user.set_password(password)
        db.session.add(user)
        db.session.commit()
        flash("Account created. Please log in.", "success")
        return redirect(url_for("auth.login"))
    return render_template("auth/register.html", form=form)


@bp.route("/forgot-password", methods=["GET", "POST"])
def forgot_password():
    if request.method == "POST":
```

gemini_ocr.py

```python
app > gemini_ocr.py > ...

def configure_gemini(app):
    """Cheap availability check (no network call) run at app startup.
    Sets app.config['GEMINI_READY'] so routes can check without re-probing."
    api_key = (app.config.get("GEMINI_API_KEY") or os.environ.get("GEMINI_AP
    if api_key and not (os.environ.get("GEMINI_API_KEY") or "").strip():
        os.environ["GEMINI_API_KEY"] = api_key
    app.config["GEMINI_API_KEY"] = api_key
    app.config["GEMINI_READY"] = bool(GENAI_AVAILABLE and api_key)


def is_available():
    return GENAI_AVAILABLE and bool((os.environ.get("GEMINI_API_KEY") or "")


def _strip_code_fences(text):
    text = text.strip()
    text = re.sub(r"^```(?:json)?\s*", "", text)
    text = re.sub(r"\s*```$", "", text)
    return text.strip()


def extract_medicines(file_path, api_key=None):
    """Send the prescription image to Gemini and ask it to extract only the
    medicine-related information. Never raises.

    Returns None on any failure (missing key/deps, network/quota error,
    unparsable response) so callers can cleanly fall back to local OCR/HTR.
    On success returns:
        {"medicines": [{"name", "dosage", "frequency", "raw_text"}, ...],
         "is_prescription": bool, "notes": str}
    """
    key = (api_key or os.environ.get("GEMINI_API_KEY") or "").strip()
    if not key:
        try:
            from app.config import _resolve_gemini_api_key
```

gemini_ocr.py

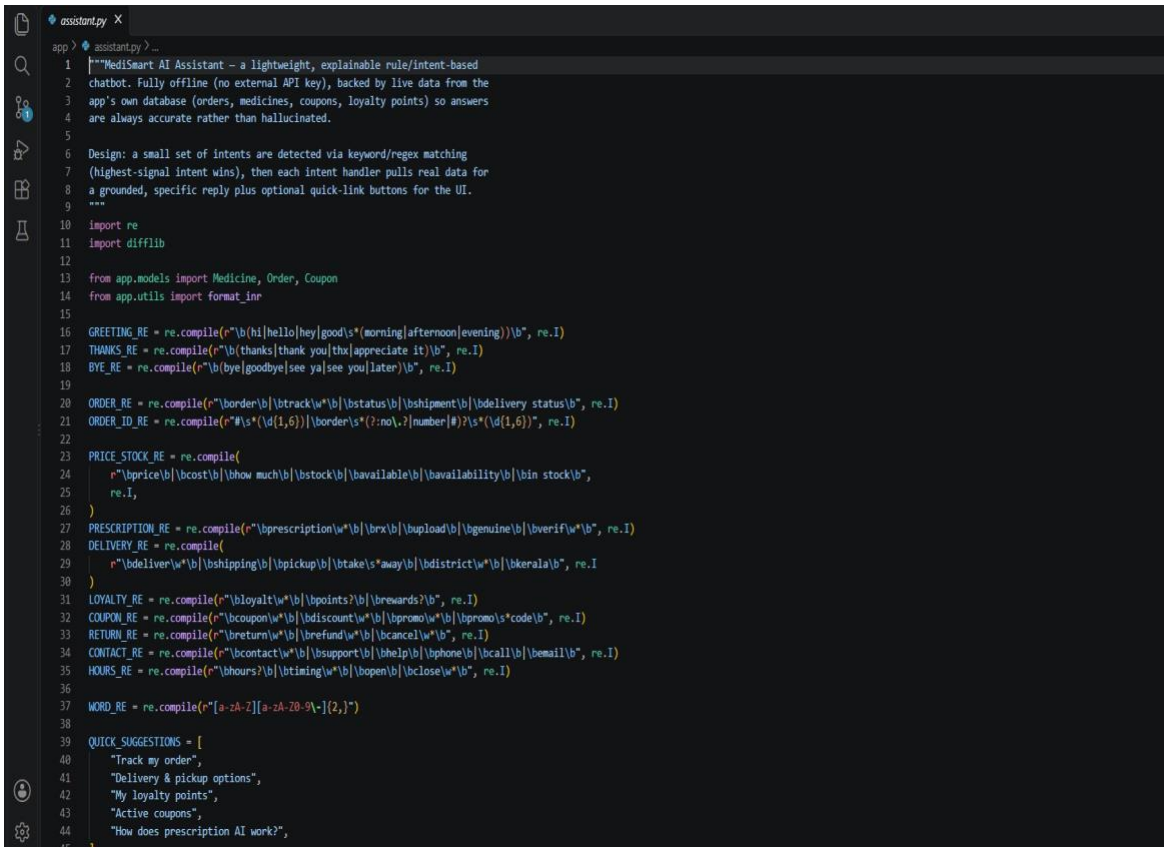
```python
app > gemini_ocr.py > ...

def extract_medicines(file_path, api_key=None):
            from app.config import _resolve_gemini_api_key
            key = _resolve_gemini_api_key()
        except Exception:
            key = ""
    if not GENAI_AVAILABLE or not key:
        return None

    ext = file_path.rsplit(".", 1)[-1].lower() if "." in file_path else ""
    mime_type = _MIME_BY_EXT.get(ext)
    if not mime_type:
        return None  # e.g. PDFs — not handled here

    try:
        with open(file_path, "rb") as f:
            image_bytes = f.read()
    except Exception:
        return None

    try:
        client = genai.Client(api_key=key)
        response = client.models.generate_content(
            model=MODEL_NAME,
            contents=[
                types.Part.from_bytes(data=image_bytes, mime_type=mime_type),
                PROMPT,
            ],
            config=types.GenerateContentConfig(
                response_mime_type="application/json",
                temperature=0.1,
            ),
        )
        raw = (response.text or "").strip()
    except Exception:
        return None
```



```python
"""MediSmart AI Assistant — a lightweight, explainable rule/intent-based
chatbot. Fully offline (no external API key), backed by live data from the
app's own database (orders, medicines, coupons, loyalty points) so answers
are always accurate rather than hallucinated.

Design: a small set of intents are detected via keyword/regex matching
(highest-signal intent wins), then each intent handler pulls real data for
a grounded, specific reply plus optional quick-link buttons for the UI.
"""
import re
import difflib

from app.models import Medicine, Order, Coupon
from app.utils import format_inr

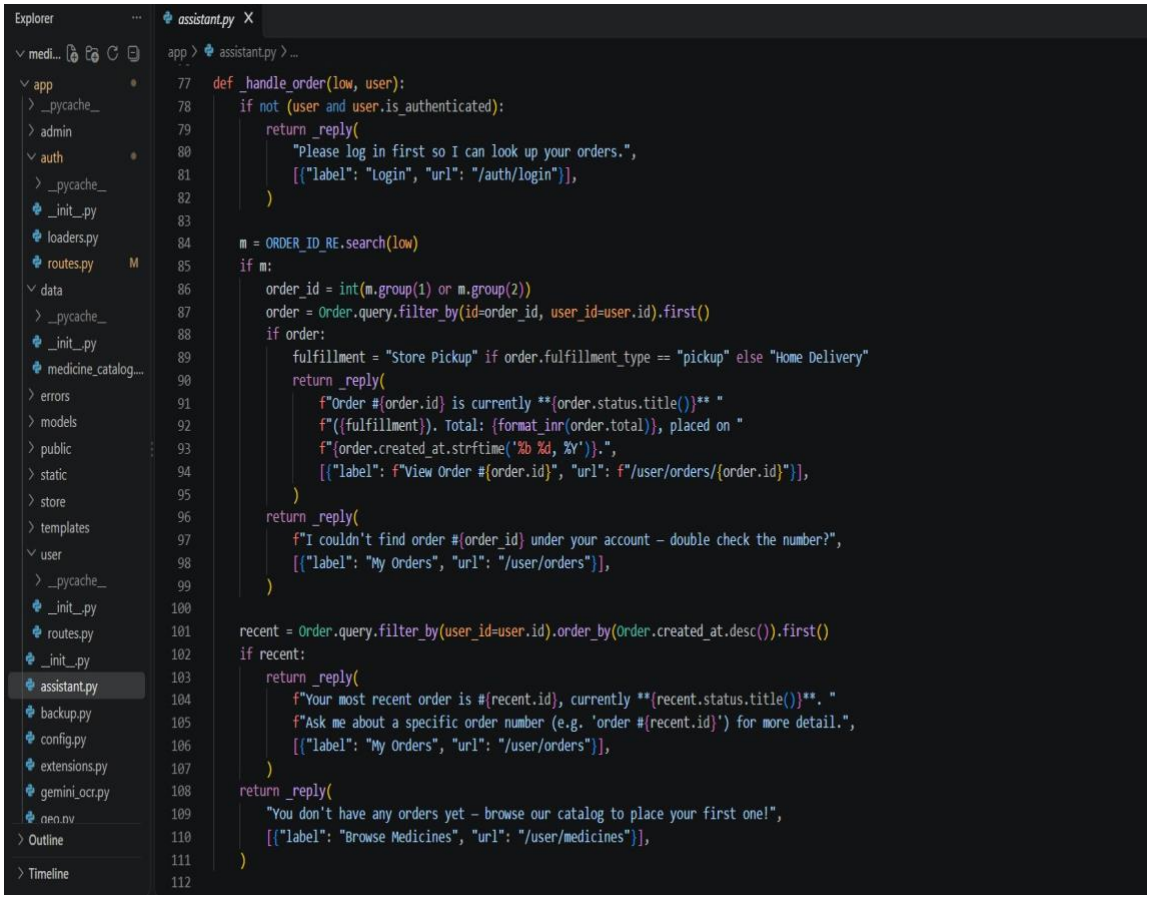
GREETING_RE = re.compile(r"\b(hi|hello|hey|good\s*(morning|afternoon|evening))\b", re.I)
THANKS_RE = re.compile(r"\b(thanks|thank you|thx|appreciate it)\b", re.I)
BYE_RE = re.compile(r"\b(bye|goodbye|see ya|see you|later)\b", re.I)

ORDER_RE = re.compile(r"\border\b|\btrack\w*\b|\bstatus\b|\bshipment\b|\bdelivery status\b", re.I)
ORDER_ID_RE = re.compile(r"#\s*(\d{1,6})|\border\s*(?:no\.?|number|#)?\s*(\d{1,6})", re.I)

PRICE_STOCK_RE = re.compile(
    r"\bprice\b|\bcost\b|\bhow much\b|\bstock\b|\bavailable\b|\bavailability\b|\bin stock\b",
    re.I,
)
PRESCRIPTION_RE = re.compile(r"\bprescription\w*\b|\brx\b|\bupload\b|\bgenuine\b|\bverif\w*\b", re.I)
DELIVERY_RE = re.compile(
    r"\bdeliver\w*\b|\bshipping\b|\bpickup\b|\btake\s*away\b|\bdistrict\w*\b|\bkerala\b", re.I
)
LOYALTY_RE = re.compile(r"\bloyalt\w*\b|\bpoints?\b|\brewards?\b", re.I)
COUPON_RE = re.compile(r"\bcoupon\w*\b|\bdiscount\w*\b|\bpromo\w*\b|\bpromo\s*code\b", re.I)
RETURN_RE = re.compile(r"\breturn\w*\b|\brefund\w*\b|\bcancel\w*\b", re.I)
CONTACT_RE = re.compile(r"\bcontact\w*\b|\bsupport\b|\bhelp\b|\bphone\b|\bcall\b|\bemail\b", re.I)
HOURS_RE = re.compile(r"\bhours?\b|\btiming\w*\b|\bopen\b|\bclose\w*\b", re.I)

WORD_RE = re.compile(r"[a-zA-Z][a-zA-Z0-9\-]{2,}")

QUICK_SUGGESTIONS = [
    "Track my order",
    "Delivery & pickup options",
    "My loyalty points",
    "Active coupons",
    "How does prescription AI work?",
```



```python
def _handle_order(low, user):
    if not (user and user.is_authenticated):
        return _reply(
            "Please log in first so I can look up your orders.",
            [{"label": "Login", "url": "/auth/login"}],
        )

    m = ORDER_ID_RE.search(low)
    if m:
        order_id = int(m.group(1) or m.group(2))
        order = Order.query.filter_by(id=order_id, user_id=user.id).first()
        if order:
            fulfillment = "Store Pickup" if order.fulfillment_type == "pickup" else "Home Delivery"
            return _reply(
                f"Order #{order.id} is currently **{order.status.title()}** "
                f"({fulfillment}). Total: {format_inr(order.total)}, placed on "
                f"{order.created_at.strftime('%b %d, %Y')}.",
                [{"label": f"View Order #{order.id}", "url": f"/user/orders/{order.id}"}],
            )
        return _reply(
            f"I couldn't find order #{order_id} under your account — double check the number?",
            [{"label": "My Orders", "url": "/user/orders"}],
        )

    recent = Order.query.filter_by(user_id=user.id).order_by(Order.created_at.desc()).first()
    if recent:
        return _reply(
            f"Your most recent order is #{recent.id}, currently **{recent.status.title()}**. "
            f"Ask me about a specific order number (e.g. 'order #{recent.id}') for more detail.",
            [{"label": "My Orders", "url": "/user/orders"}],
        )
    return _reply(
        "You don't have any orders yet — browse our catalog to place your first one!",
        [{"label": "Browse Medicines", "url": "/user/medicines"}],
    )
```

CHAPTER 11

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